

Waves, Sound and Light

Grade 10 physics

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30 Years of Excellence
1987 - 2017

Heartbeat

Flick in hose pipe

What is a pulse?

- A single disturbance that moves through a medium.

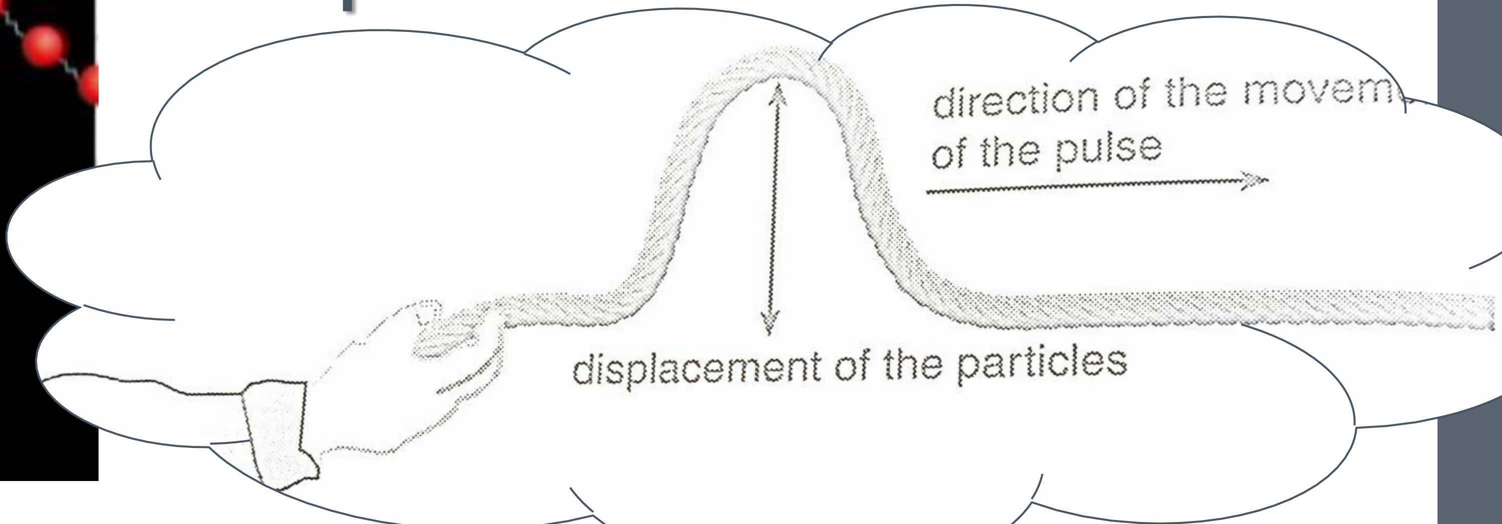
Stone in water

Other?

moving

Transverse pulse:

A pulse within which the displacement of the particles of the medium is perpendicular to the direction of the movement of the pulse.



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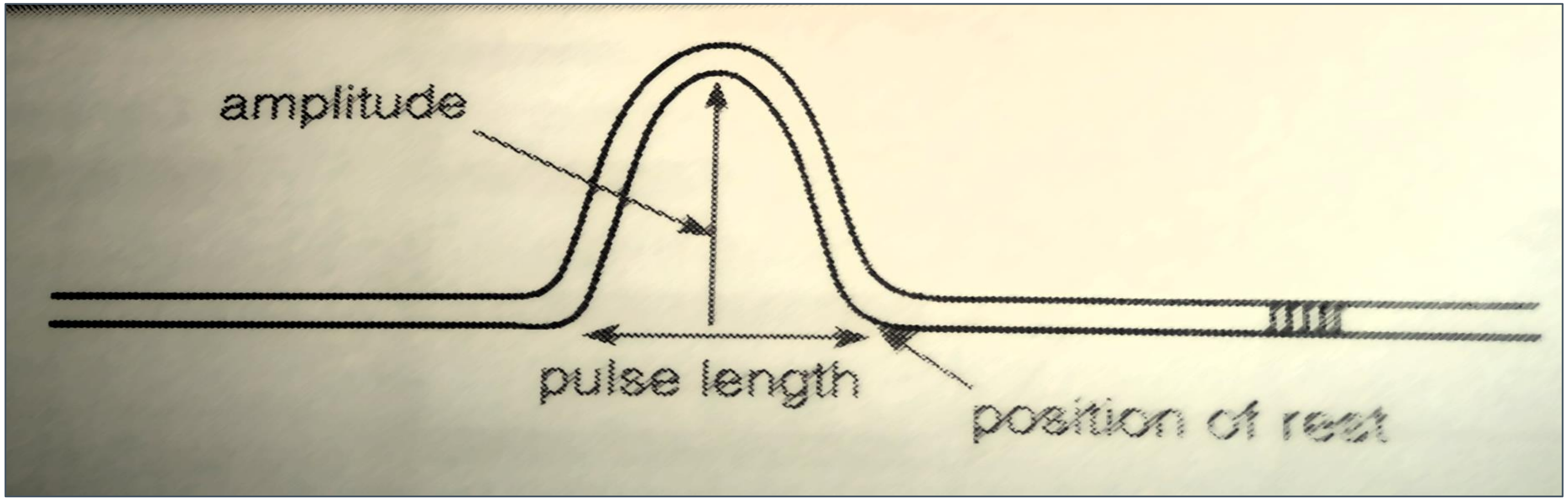
› **Amplitude:**

– Maximum displacement of a particle from its position of rest (equilibrium).



› **Pulse length:**

– Distance between the start and end of a pulse.

› **Speed of a pulse:**

Distance (m)

$$\boldsymbol{v} = \frac{\boldsymbol{d}}{\Delta \boldsymbol{t}}$$

$(\mathrm{m}.s^{-1})$

Change in time (s)

LET'S DO SOME MATHS FIRST

1. Solve for x

$$2x - 5 = 7$$

2. Solve for d

$$10 = \frac{20}{d}$$

3. Solve for s

$$10 = \frac{20 - s}{5}$$

HOW TO USE EQUATIONS IN PHYSICS

Transverse Phase

Type of Measurement	Unit/ Abbreviation	Tool Used
Length	Meter, m	Meter Stick, measuring tape
Mass	Kilogram, kg	Electronic scale or balance
Volume	Liter, l	Graduated Cylinder
Time	Seconds, s	Stopwatch or clock
Temperature	Celsius ° C, Kelvin °K	Thermometer

ation.

use.

8. Find the values that you are looking for.
9. ANSWER the question.

Transverse Pulses

› **Example 1**

It takes 0.2s to produce a pulse. The distance covered is 300mm. Calculate the speed of the pulse.

› **Example 2**

› The speed of a pulse is 0.032m.s^{-1}. Calculate the distance that the pulse will cover in 2 minutes.

Exercise 1 pg. 14

Interference

- Whenever two pulses in the same medium meet, they will interact with each other. This interaction is called **interference**.
- Definition: **The overlapping of two pulses that are at the same point at the same time.**
- The sum of the amplitudes of the two pulses when they interact with each other is known as superposition.

Interference

What can you conclude about interference?
IOW What impact did it have on the two pulses?

Types

Constructive

When 2 pulses meet each other on the same side of the rest position.
Results in a pulse with a greater amplitude.

Destructive

When 2 pulses meet each other on opposite sides of the rest position.
Results in a pulse with a greater amplitude.

Afterwards: Continue in original directions with original amplitudes.

Rest position – “Equilibrium”

Interference

- **Superposition**
  - The sum of the amplitudes of the two pulses which overlap when they are at the same place at the same time.

- **Constructive Interference**
  - When 2 pulses meet each other on the same side of the rest position. The resulting amplitude is larger.

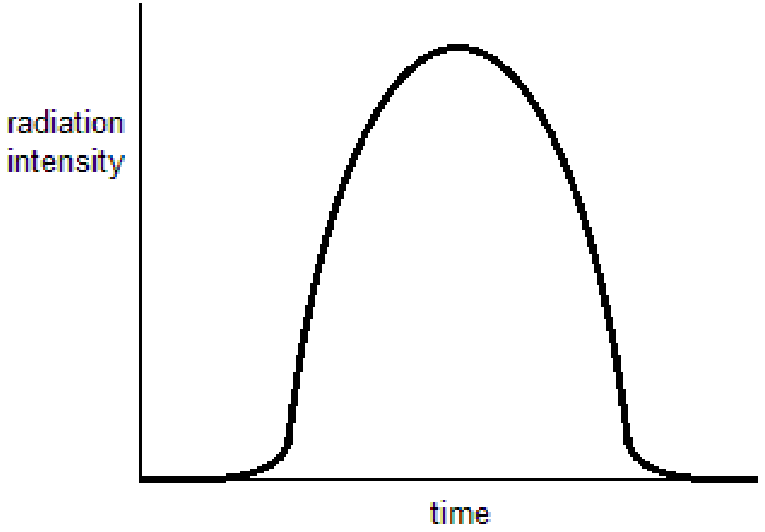
- **Destructive interference**
  - When 2 pulses meet on opposite sides of the rest position. The resulting amplitude is smaller.

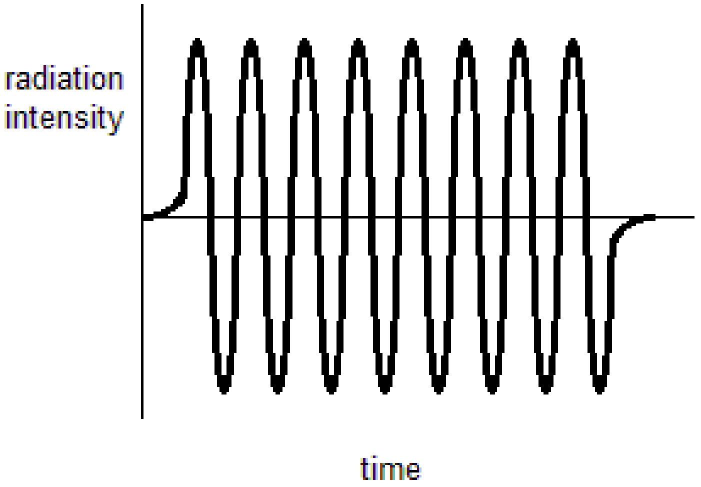
Exercise 2



Transverse Waves

The difference between a **pulse** and a **wave**



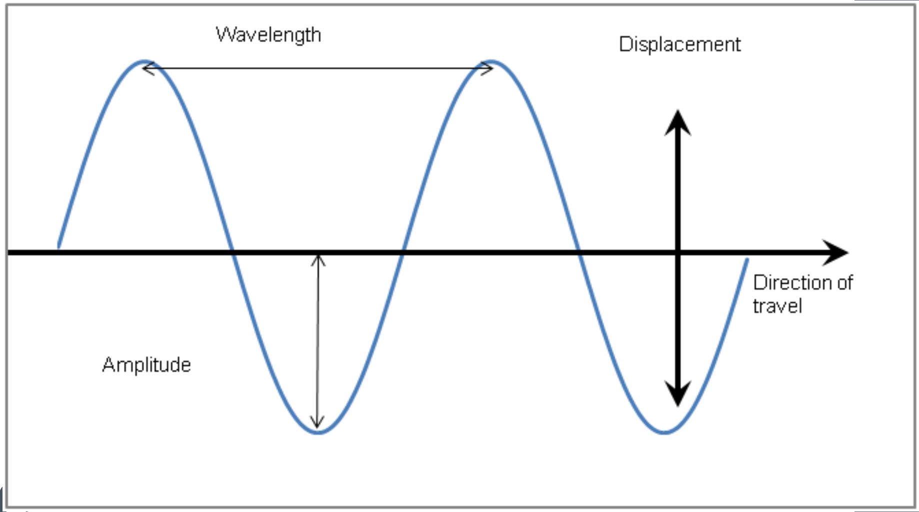


A single disturbance that moves through a medium

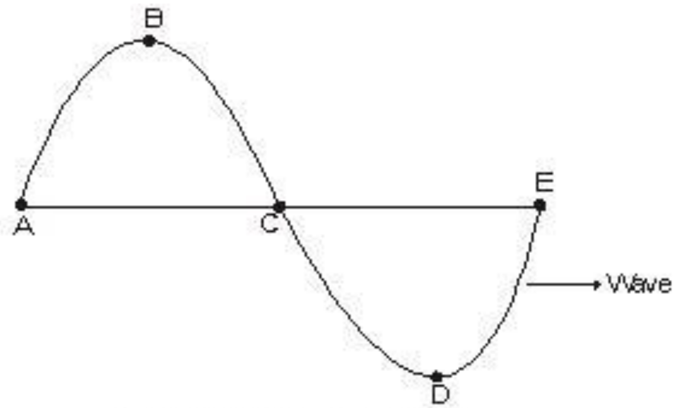
A repletion of pulse.

Definition

Transverse Wave

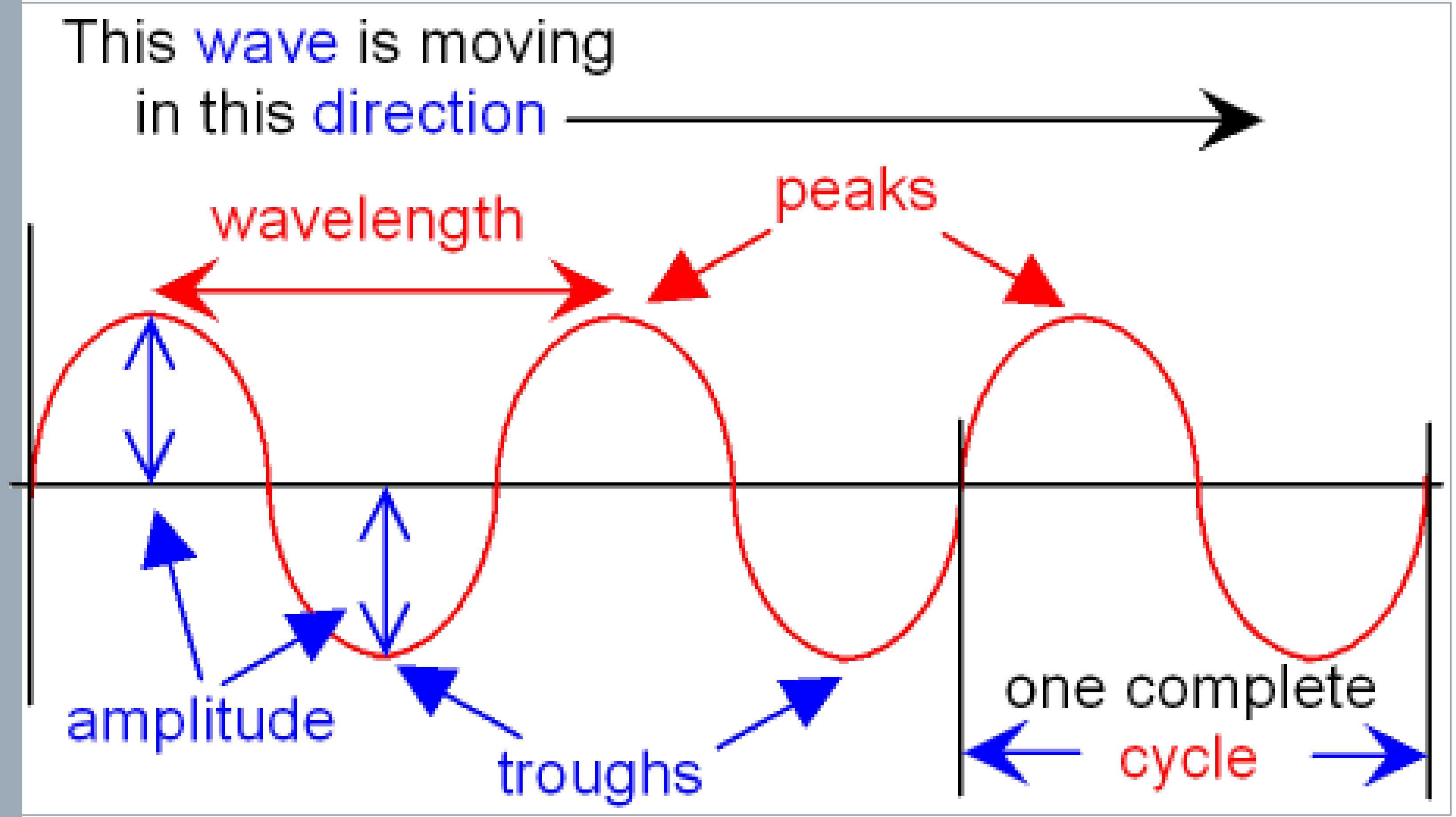


- Repetition of transverse pulses
- A wave in which the disturbance of the medium is **perpendicular** to the direction of the wave.
- Every up and down movement past the position of equilibrium is one complete oscillation.
- **Oscillation**: 2 consecutive pulses – same amplitude and opposite signs.



Wave showing one oscillation

Illustration

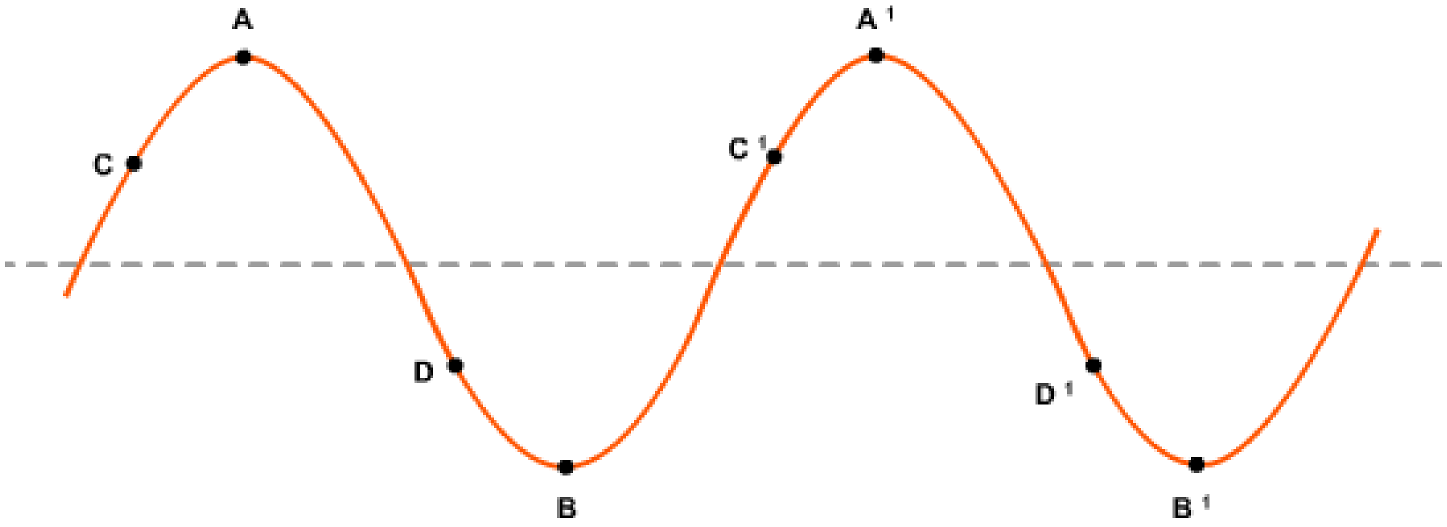


Amplitude:
Maximum displacement from start to rest

Crest:
Highest point of a wave, at it's maximum displacement from equilibrium

Trough:
Lowest point of a wave, at it's maximum displacement from equilibrium in opposite direction

Phase



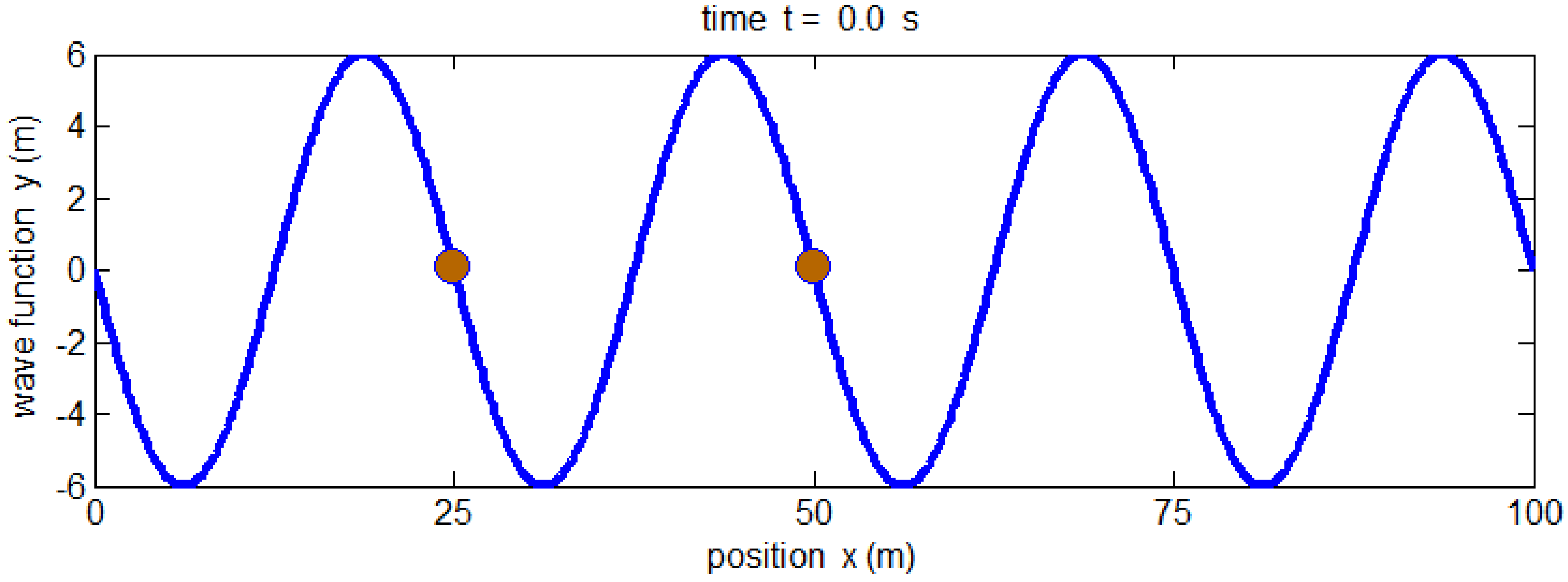
Points in phase

Points doing exactly the same thing at the same time and are equal distances from the equilibrium position in the same direction.

Points out of phase

Points **not** doing exactly the same thing at the same time.

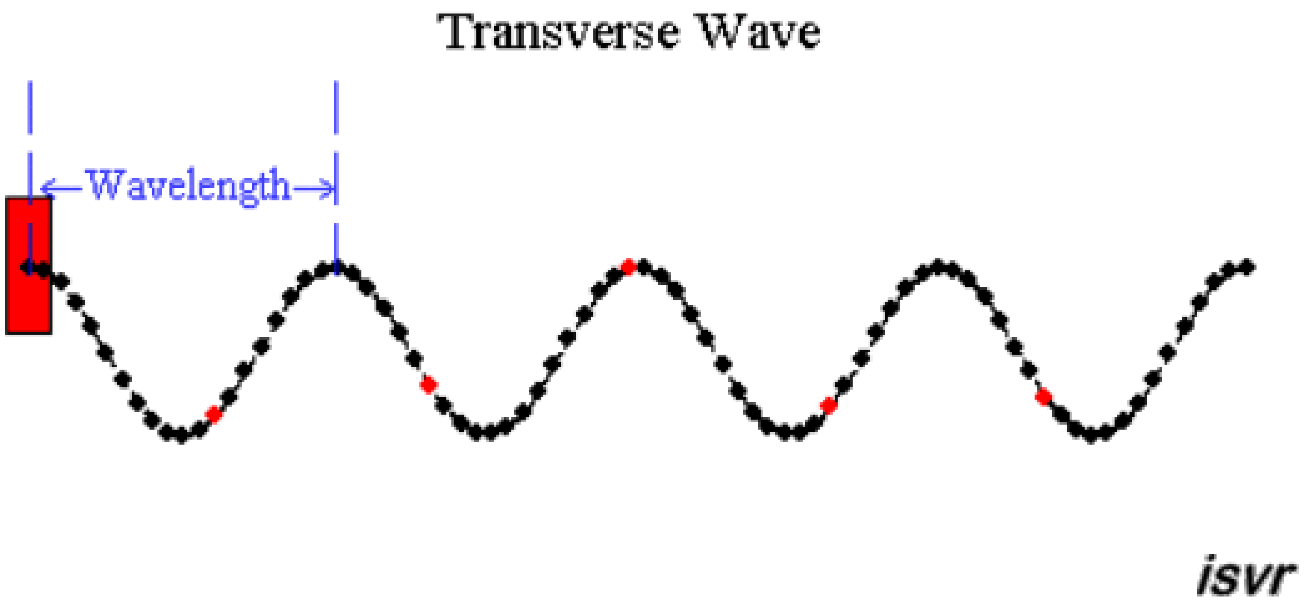
Frequency



- **Definition: The number of pulses (oscillations) that pass a point per second**
- **Unit: Hz**
- $\dfrac{number\ of\ waves}{time}$ **or** $f = \dfrac{1}{T}$

A frequency of 4Hz means that 4 waves pass a point per second.

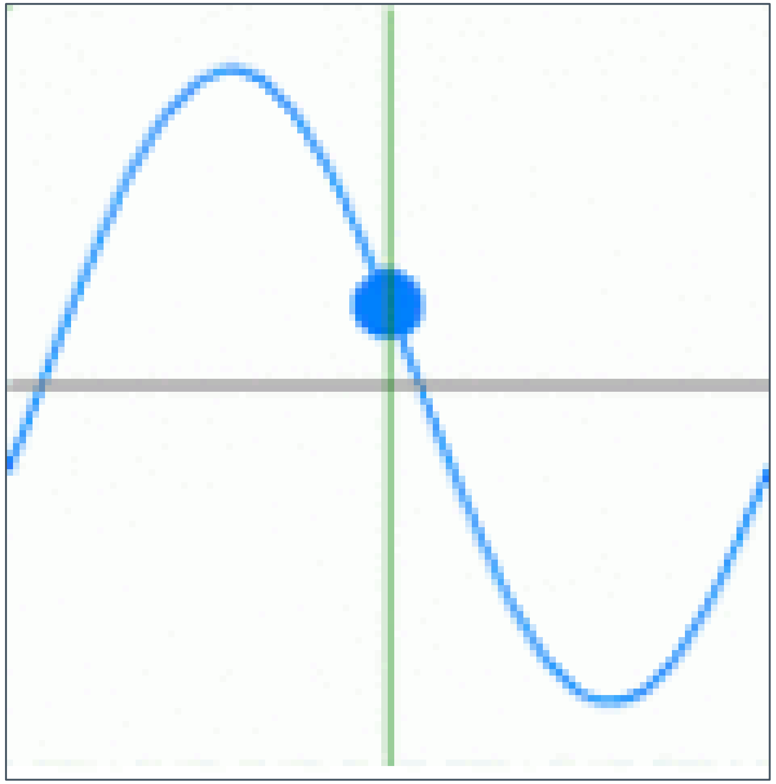
Wavelength



- **Definition: The distance between two consecutive points in phase.**
- **Unit: m**

A period of 4s means that every 4s, a complete waves moves past a fixed point.

Period



- Definition: The time taken for one complete wavelength to pass a point.
- Unit: s

Period & Frequency relation

$$f = \frac{1}{T} \qquad or \qquad T = \frac{1}{f}$$

Wave speed

- The speed of a wave is the distance that a wave covers in 1 second.
- Represented by 2 equations:

$$v = \frac{d}{\Delta t}$$

v ($m.s^{-1}$)

d: Distance (m)

Δt: Change in time (s)

$$v = f\lambda$$

v ($m.s^{-1}$)

f: Frequency (Hz)

λ: Wavelength (m)

Examples

EXAMPLE 1

A water wave comes into the harbor at a speed of 1.5 and has a wavelength of 200cm. Calculate the frequency with which the wave hits the harbor wall.

- **EXAMPLE 2**
- Five waves cover a distance of 10cm in 2 seconds.
- Calculate

a) the speed of the waves

b) The frequency of the waves

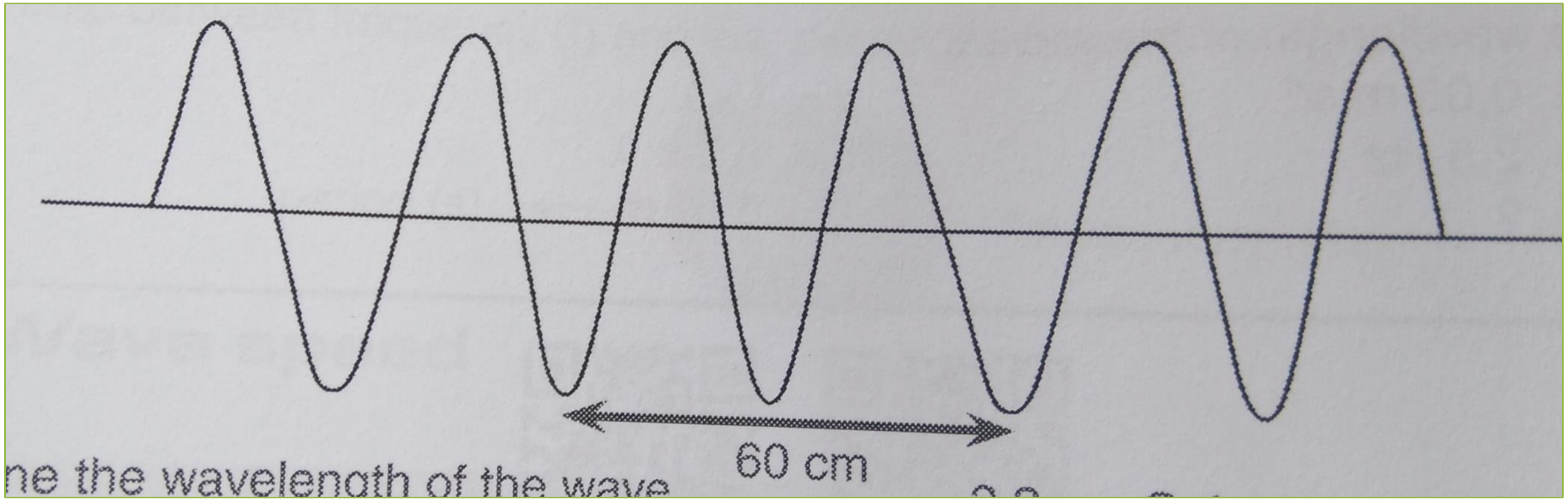
c) The period of the waves

d) The wavelengths of the waves

Exercise 3

Do question 1 and 2

A transverse wave is produced in a rope. The wave is represented below



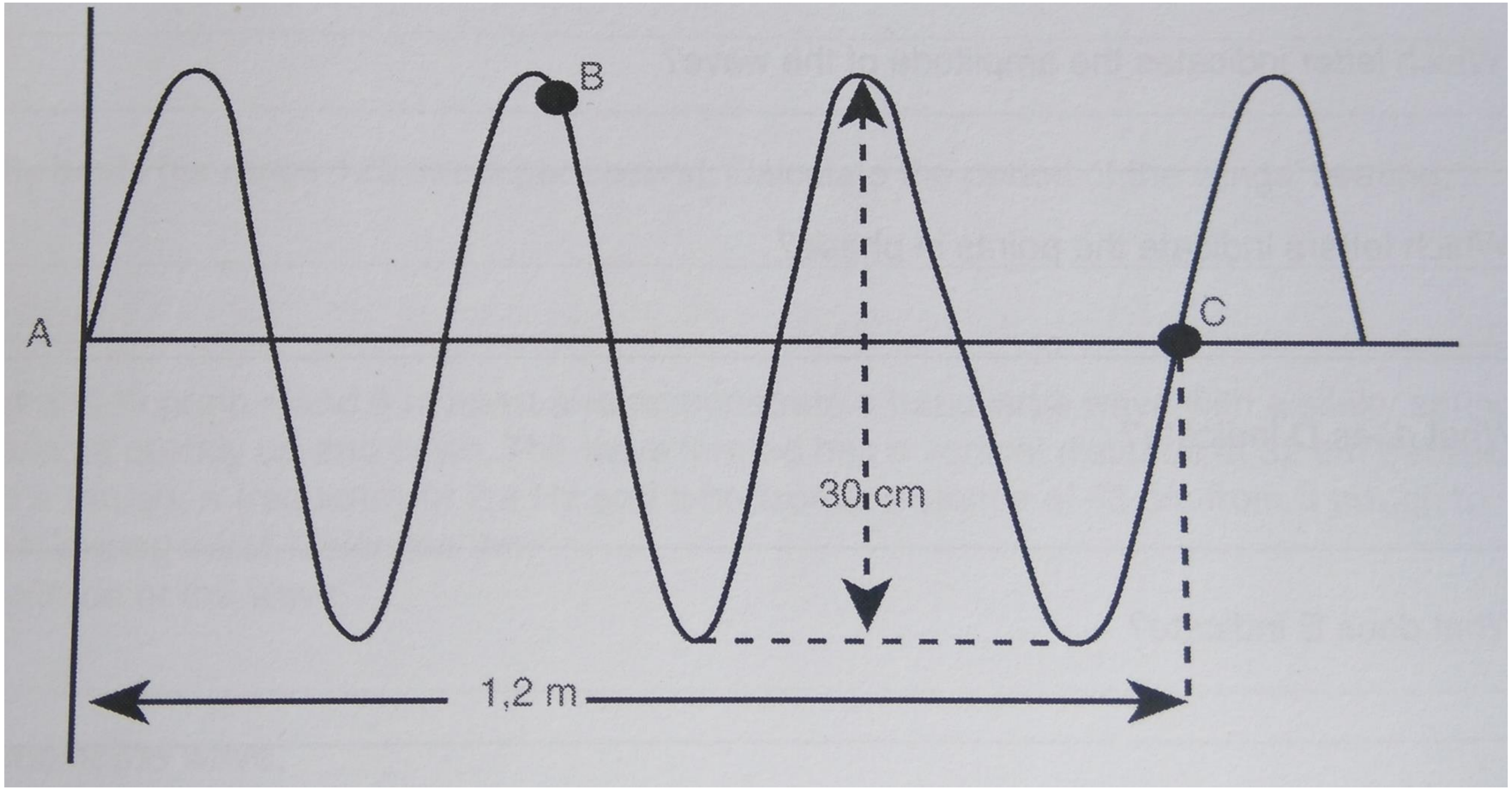
3.1 Determine the wavelength of the wave

3.2 Calculate the wave speed if the wave has a frequency of 4Hz.

3.3 Calculate the period of the wave.

Exercise 3

The following diagram represents a wave with a frequency of 10Hz.



5.1 Calculate the wavelength of the wave

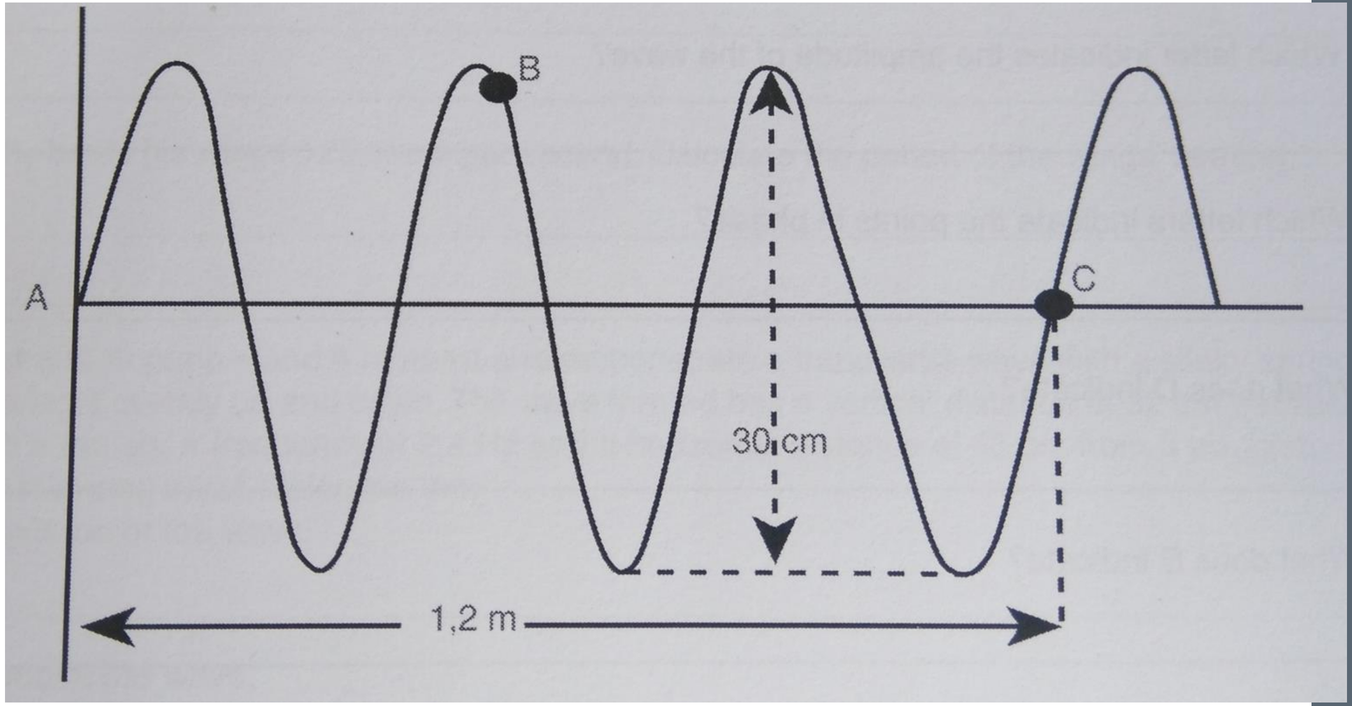
5.2 What is the amplitude of the wave?

5.3 In which direction is point B moving?

5.4 Calculate how long it will take for 4 waves to move past point C.

5.5 Calculate the speed of the waves


Transverse Waves

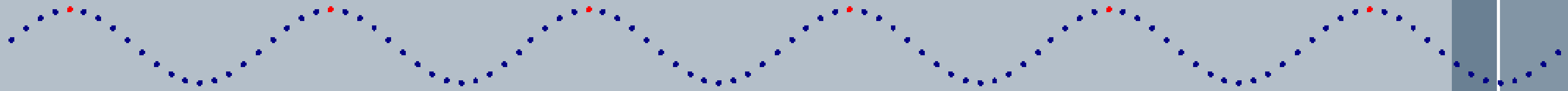

A
B
C
30 cm
1,2 m

Exercise 3 pg. 27

HOMEWORK

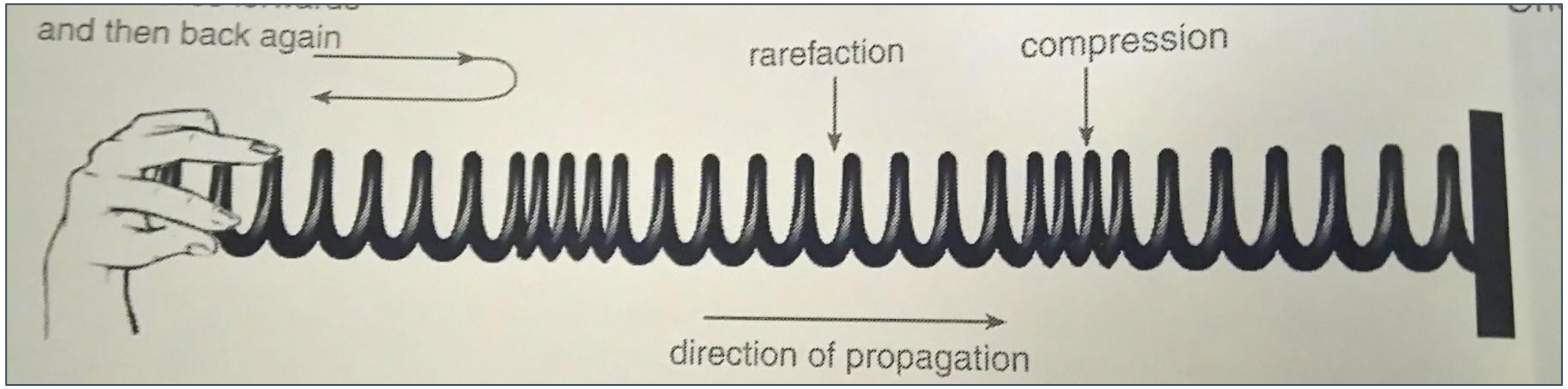


Longitudinal Waves



Slinky spring

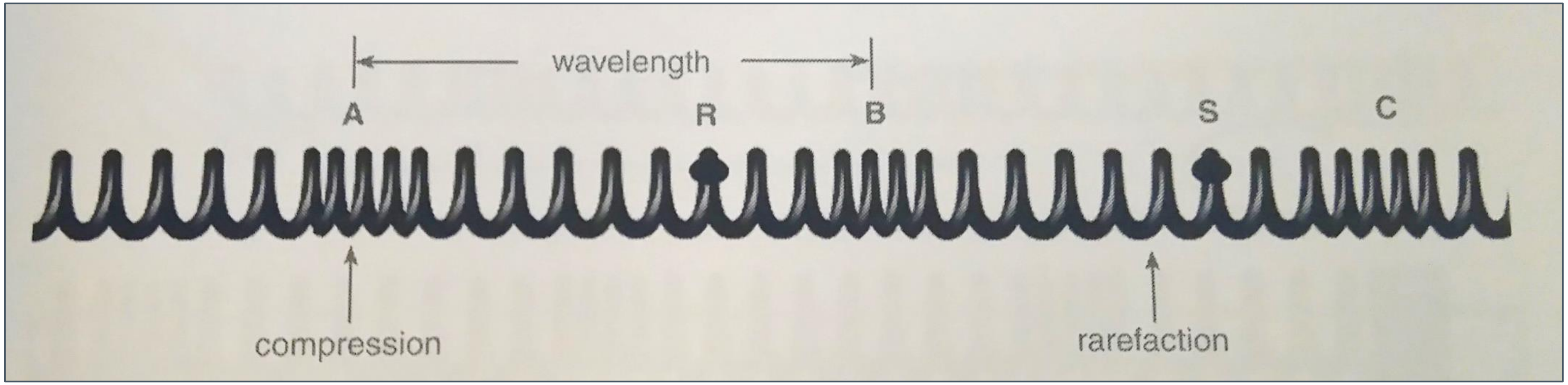
- Forward and backward motion of hand = 1 pulse
- Spirals are close together = **COMPRESSION**
- Behind compression – stretched part = **RAREFACTION**



$$> energy => displacement$$

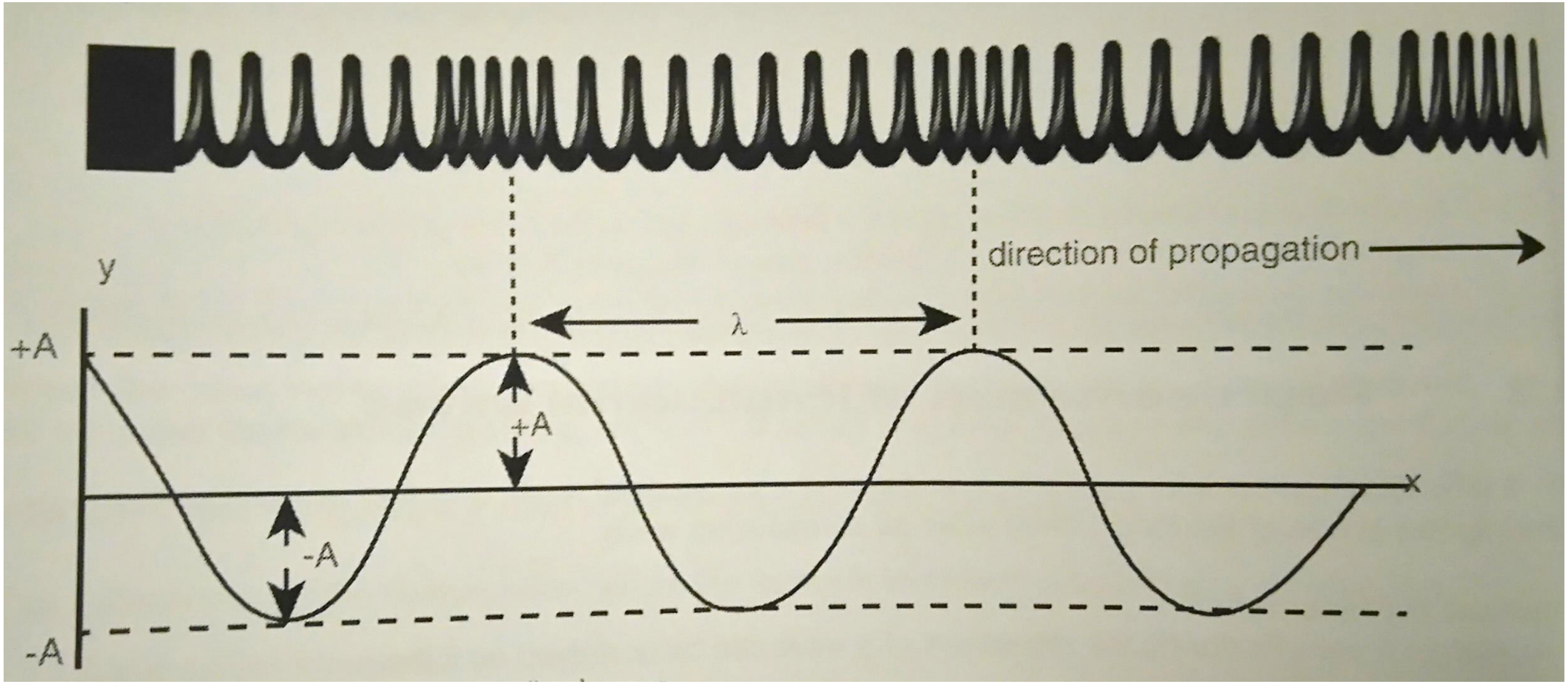
- **Definition of longitudinal wave:**
- A wave in which the particles of the medium move parallel to the direction of the waves.

Representing a longitudinal wave



- A and B are in phase = the distance between A and B = wavelength
- Maximum distance that a particle has from its rest position (equilibrium) = amplitude.

- MAXIMUM DISTANCE FROM REST POSITION = Centre of compression/rarefaction.



- **Wavelength**: Distance between 2 compressions/ 2 rarefactions
- **Period**: Time that one complete waves takes to move past a fix point per second.
- **Frequency**: Number of waves that move past a point per second.

Equations

$$\mathbf{f} = \frac{1}{T}$$

$$\mathbf{T} = \frac{1}{f}$$

Frequency (Hz)

$$v = \frac{d}{\Delta t}$$

Distance (m)

$(\mathrm{m}.s^{-1})$

Change in time (s)

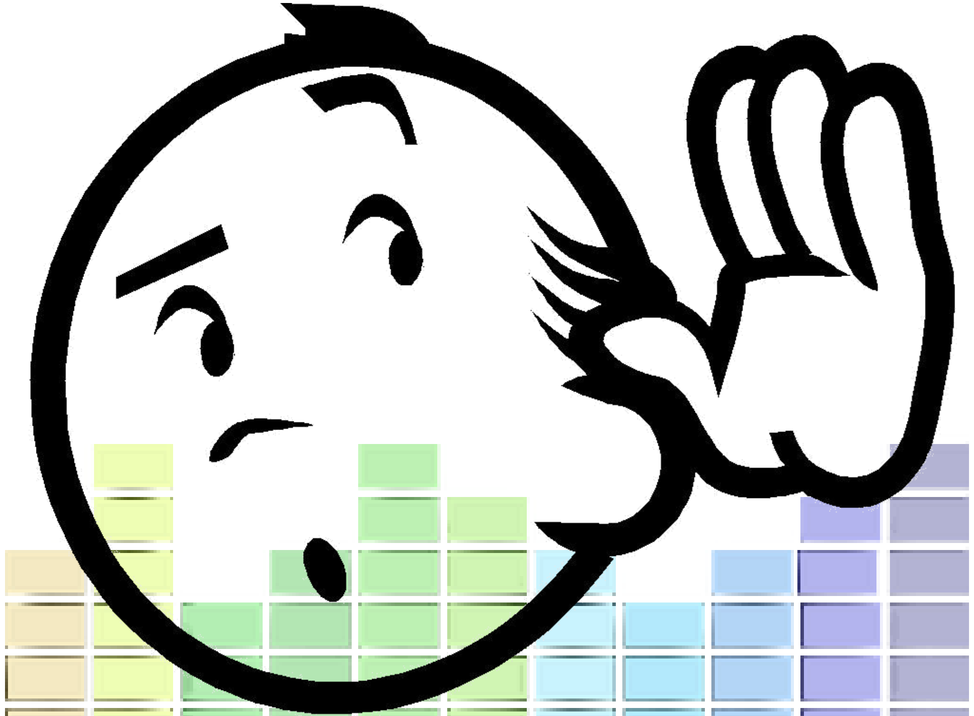
$$v = f\lambda$$

$(\mathrm{m}.s^{-1})$

Wavelength (m)

Sound

As a longitudinal wave



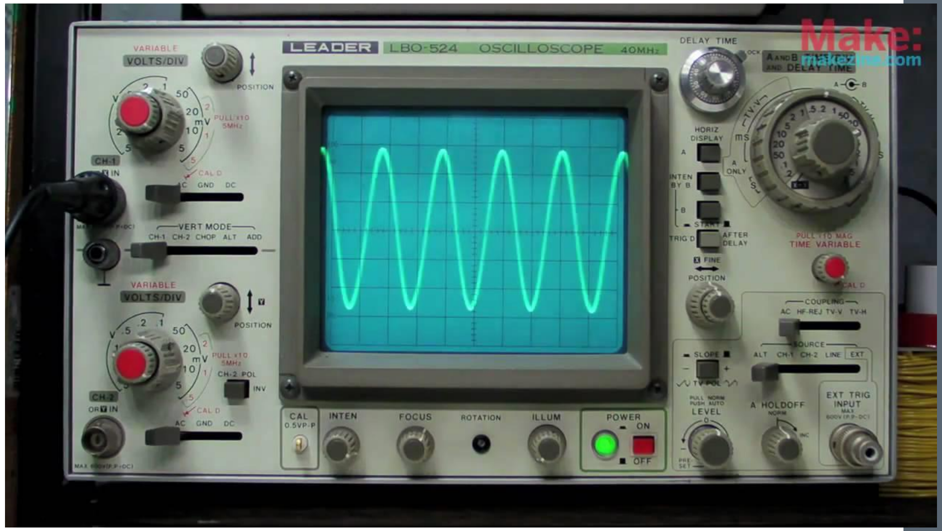
- Sound needs a **medium** through which to travel.
- The **compressions and rarefactions** move towards a persons ear where the sound can be heard.
- The **number of vibrations** per second is the frequency of the wave.

OSCILLOSCOPE

- We use an **oscilloscope** to view these longitudinal waves that are displayed as sound waves.
- The **crests** on the transverse wave refers to the **compressions** in the longitudinal wave.
- The **trough** on the transverse

wave refer to the **rarefactions**

on the longitudinal wave.

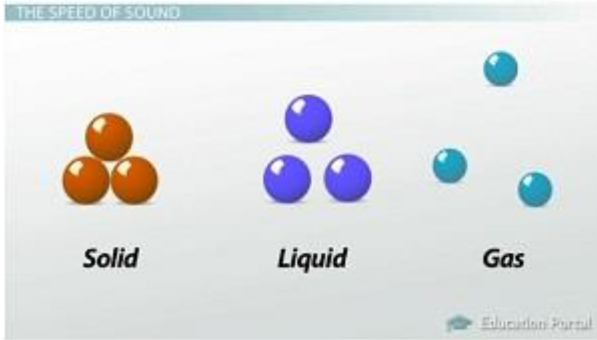


SPEED OF WAVES

- 2 factors that determine the speed of a waves:

  - Elasticity of the medium (how stiff the medium is)
    - the **greater** the elasticity – the **faster** speed can travel. E.g. sound travels faster through steel than through rubber and faster in solids than liquids or gases.

  - Density of the medium
    - The **lower** the density, the **faster** the sound travels. Sound travels faster in hot air or water (low density) than in cold air or water.



Exercise 4 and 5 pg. 42-51

HOMEWORK

Properties of sound

Property #1

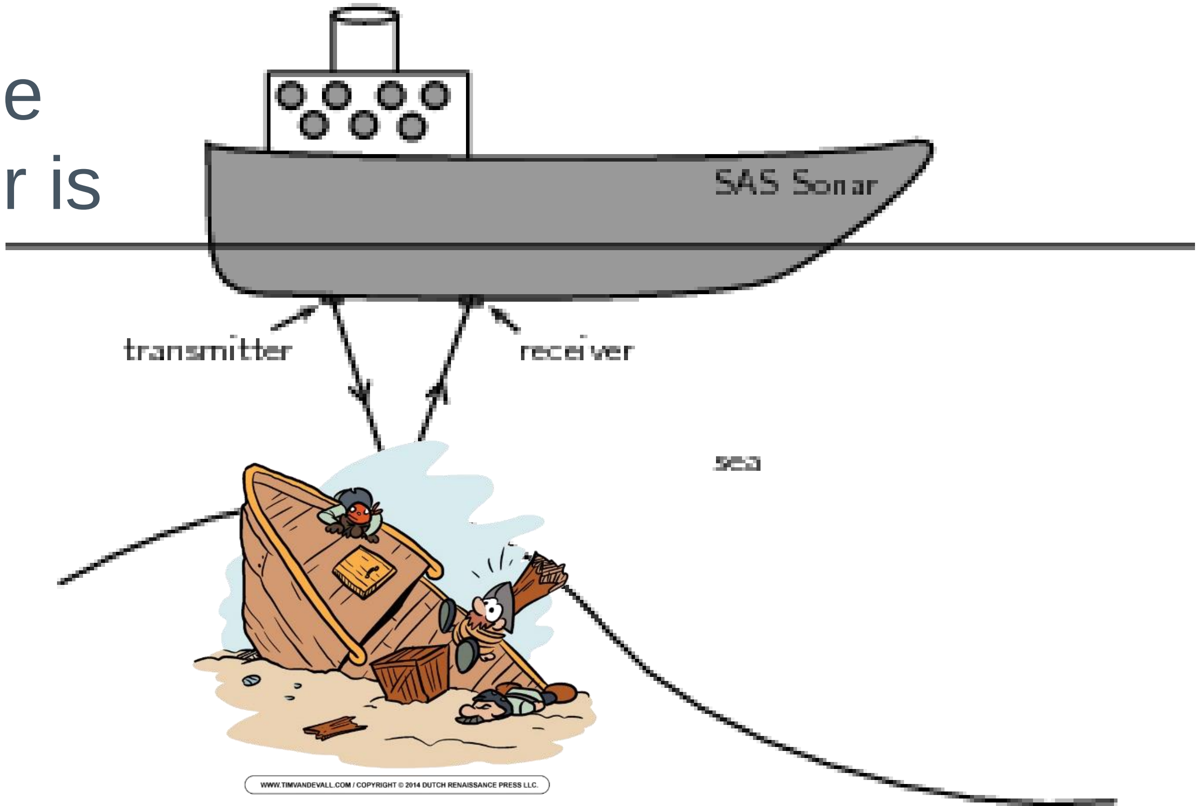
Reflection

- Most **well-known property** of sounds.
- When you talk in a furnished room – sound is **absorbed** by the carpets, furniture, curtains, wall hangings etc.
- If these were not here – your voice would reflection off the walls. This reflection is called an **echo**.

- Objects absorb energy. So when a sound is reflected by an object, it will always be **softer** since part of the **energy** has been absorbed by the reflecting object.

Calculations with echo's

A boat transmits a sound wave, and 5s later it registers its reflection. Calculate the depth of the shipwreck, if the speed of sound in sea water is $1480 m.s^{-1}$.

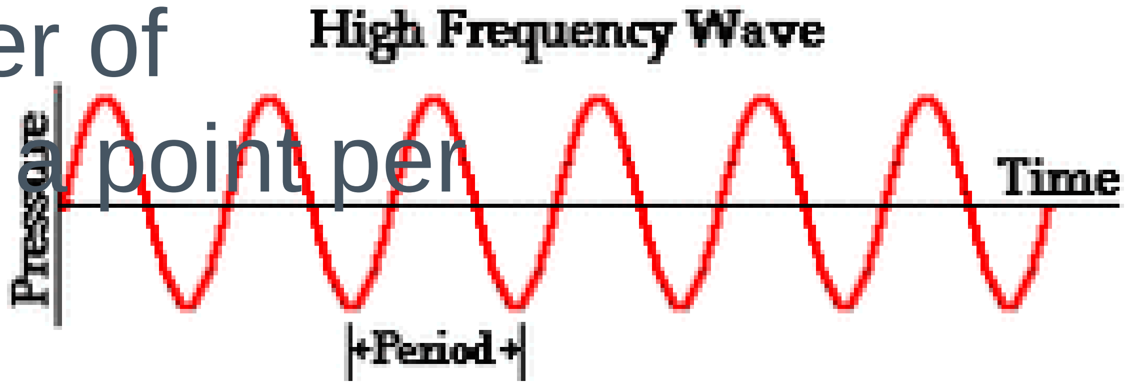


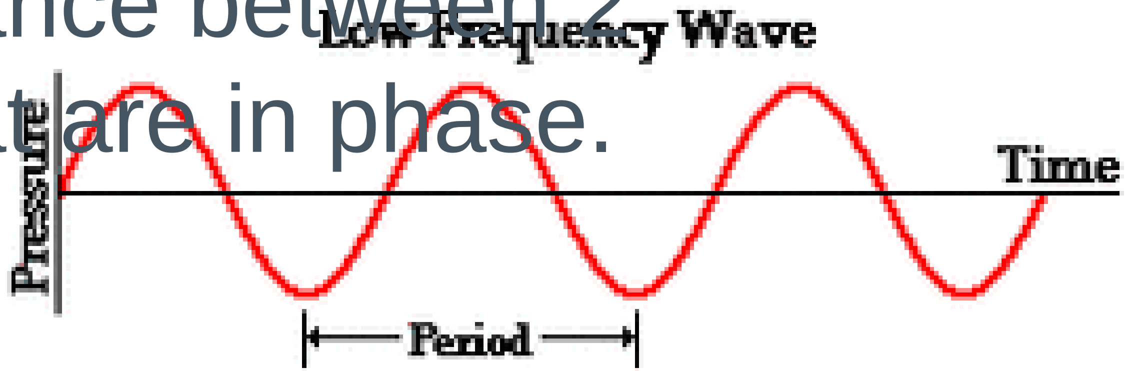
The relationship between frequency and wavelength

Frequency: The number of wavelengths that pass a point per second.

Wavelength: The distance between 2 consecutive points that are in phase.

- A **smaller wavelength** means a **greater frequency** (more waves can pass a point per second).
- A greater wavelength means a smaller frequency.





Write down

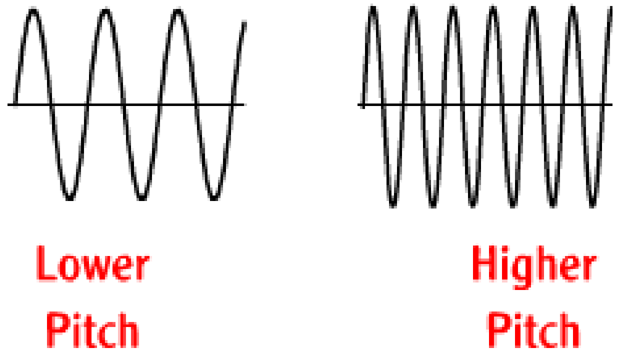
Property #1

› **PITCH** has everything to do with **FREQUENCY**.

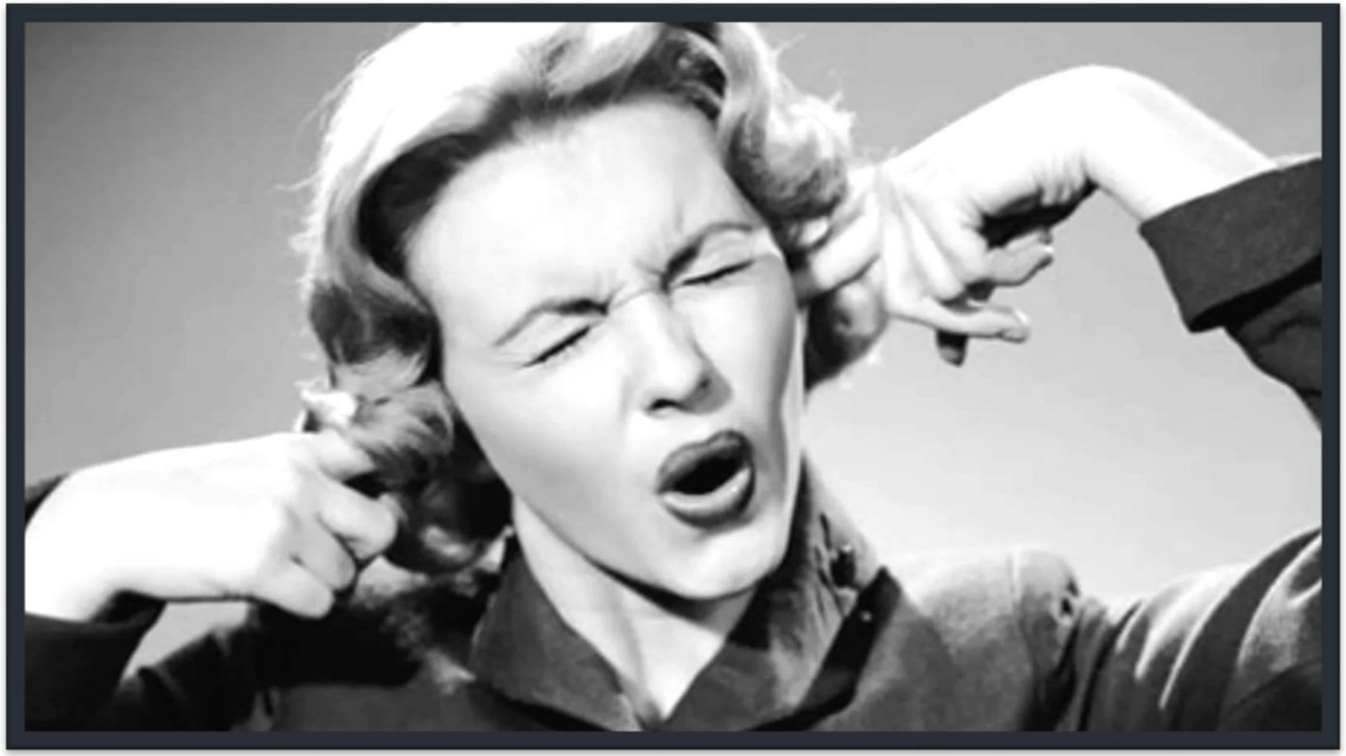
- Pitch is **how high or low the note sounds**.

*Low frequency = low note = low pitch - **Nkanyezi's voice**

*High frequency = high note = high pitch - **Terease's voice**



Property #3



Volume

› **VOLUME** has everything to do with **AMPLITUDE**.

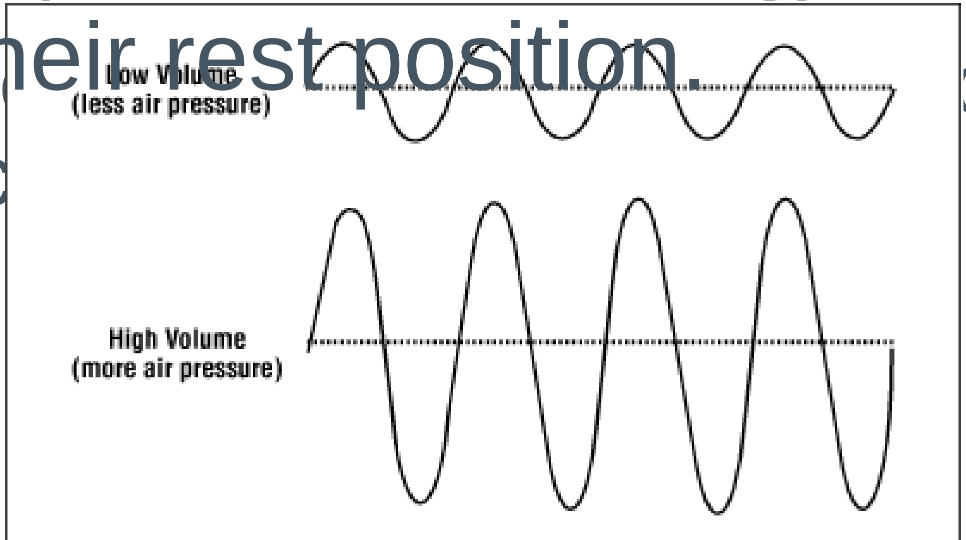
*__Low__ volume = **small** amplitude = **soft** sound - **whisper**

*__High__ volume = **large** amplitude = **loud** sound – **scream**

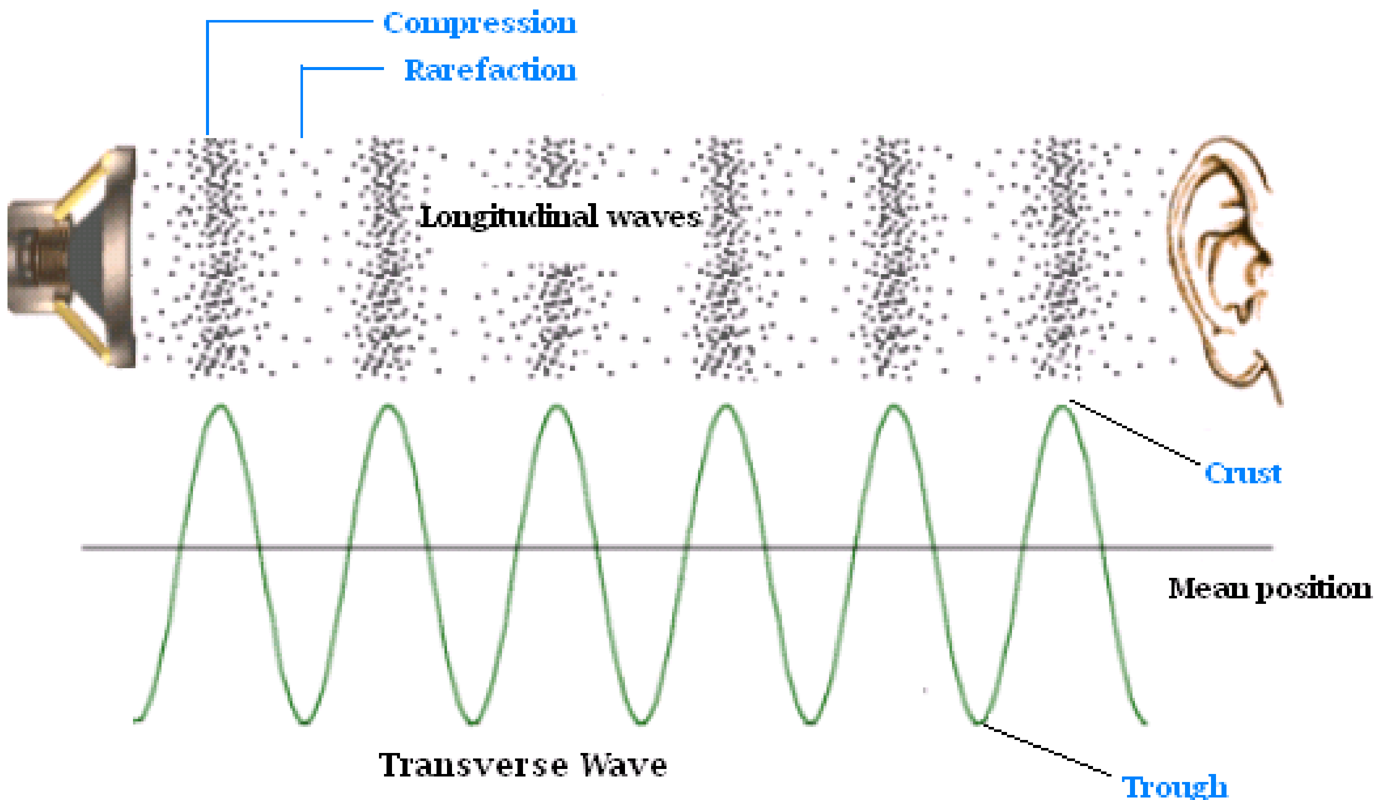
Amplitude: The maximum displacement of particles in a wave from their rest position.

*Amplitude is proportional to the enrgy of the wave.

*Sensitivity of th … on the loudness that is experienc…



- **Louder** sound – **greater** degree of compression and rarefaction
- Sound is measured is **decibels** (dB)
- Frequency (i.e. pitch) and wavelength **remains the same** – only the amplitude changes.

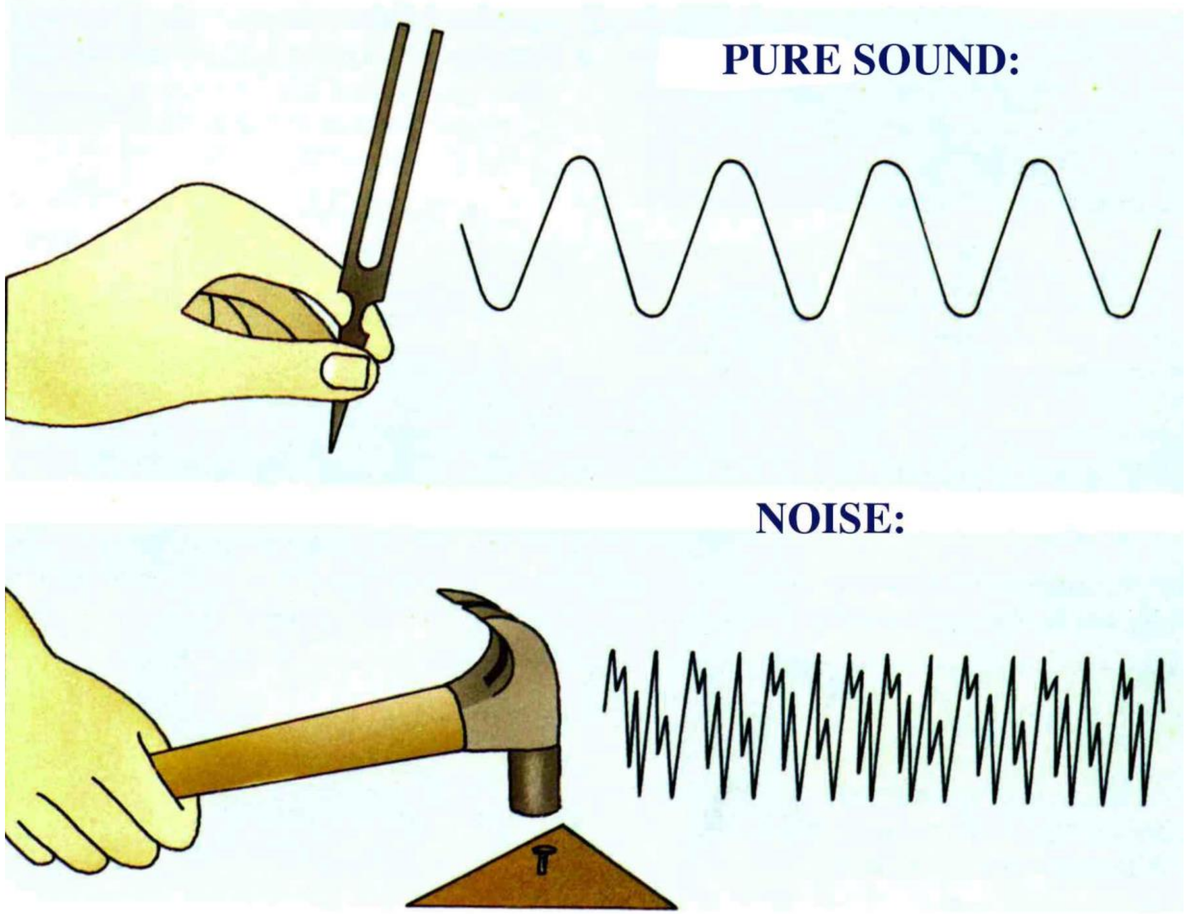


Property #4

Quality

- A **pure sound** gives a **regular pattern** e.g. Sound from a tuning fork.

- An **impure** sound gives an **irregular pattern**. E.g. sound from a vuvuzela.



Ultrasound

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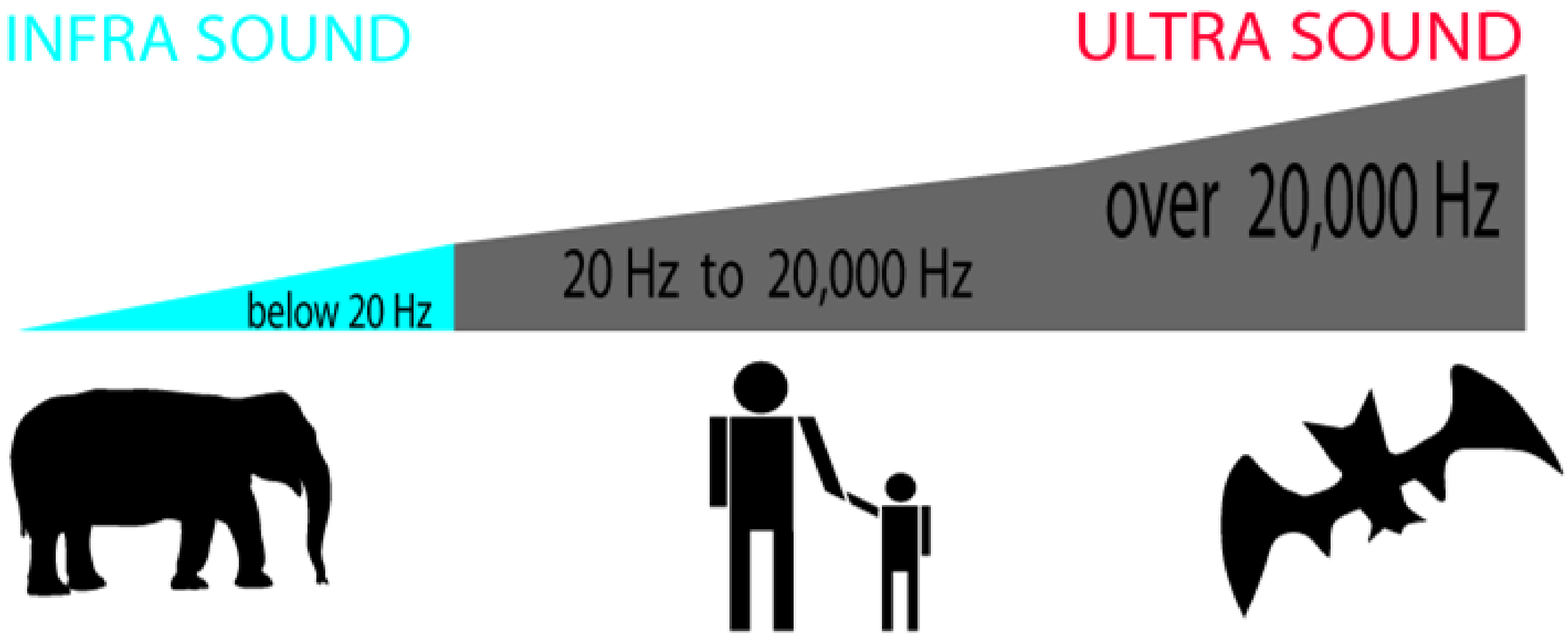
- We can hear sounds between **20Hz and 20 000Hz**.
- Any frequency above 20 000Hz is known as **ultrasound**.



In order for a sound wave to reflect off an object – the **object must be bigger than the wavelength** of a sound.

This is why bats send out notes with a high frequency and short wavelength.

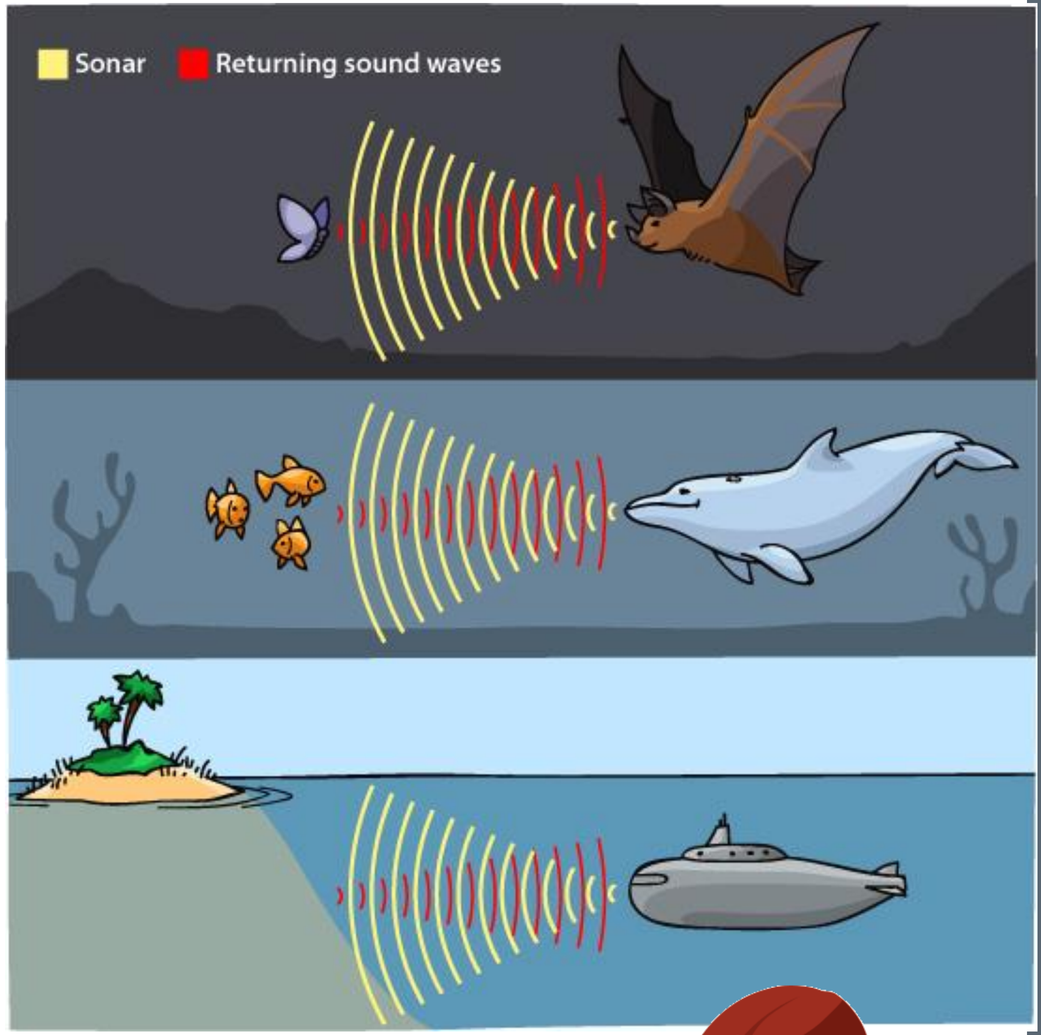
Uses of ultrasound

Dolphins and **bats** often use ultrasound to hunt their prey.

– **Ships.** {Sonar}SONAR

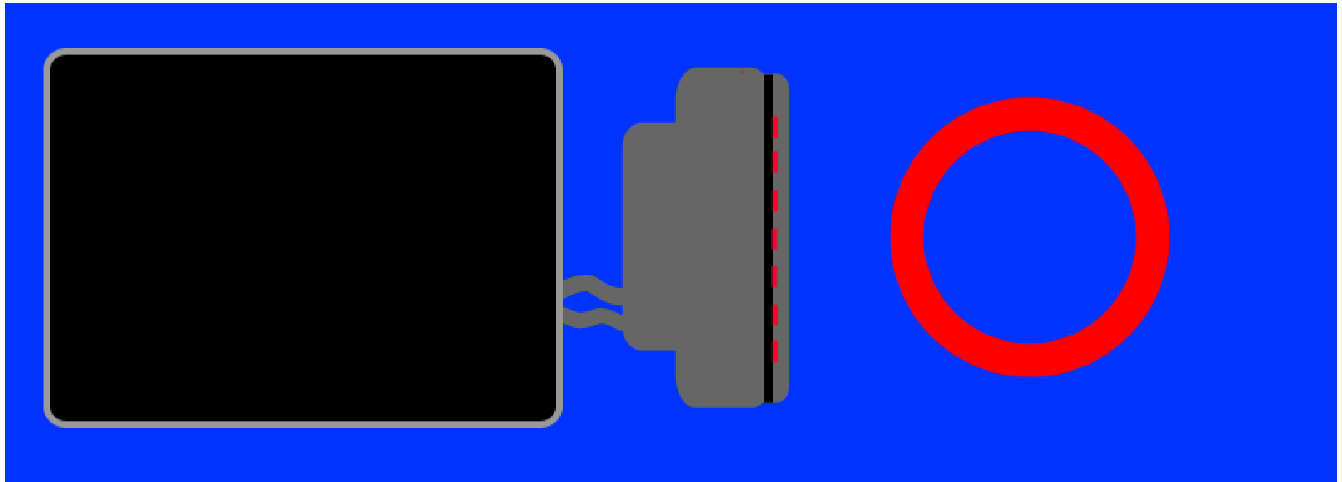
- – Transmitter **sends a sound wave** into the water.
- – Sound wave **reflects** off the bottom of the sea/object and returns to the transmitter on the ship.
- – **Time** taken for the sound wave to return is recorded and used to calculate the **depth** of the sea/object.

› Used to **track** shipwrecks and schools of fish



Ultrasound in the medical world

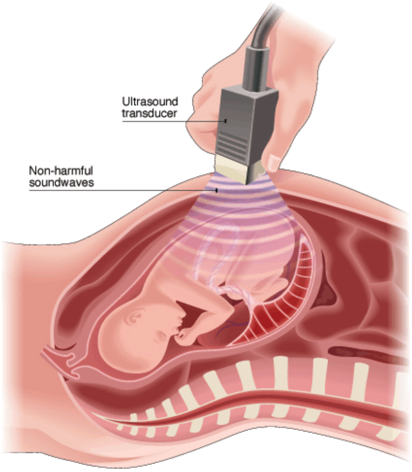
- Ultrasound machines have replaced X-ray machines.
- When ultrasound waves are sent through **tissue**, the waves are partially **reflected**, partially **transmitted** and partially **absorbed** at the interface between tissues of different densities. E.g. bone and muscle or fat etc.
- The **reflected** waves are picked up by a **receiver** which then sends them to the **computer** that converts them into an **image.**



› Pregnancy

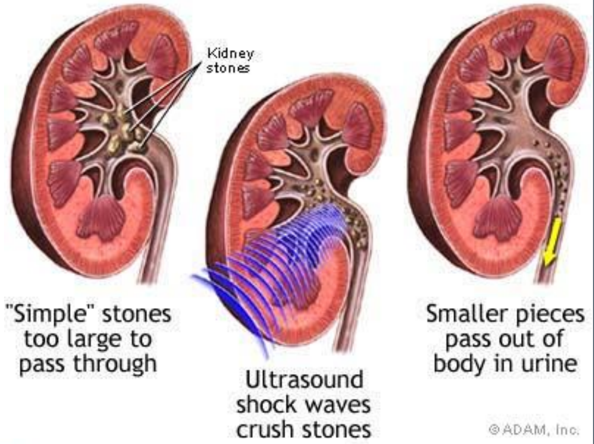
– Sonar of foetus.

– Location, size, organs, more than 1 etc.



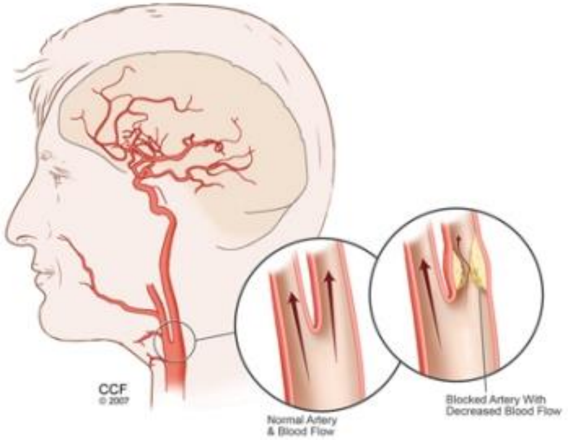
› Treatment

– Kidney stones. Ultrasound can break them up so that the patient can pass them without too much pain for having to undergo an operation.



› Diagnosis

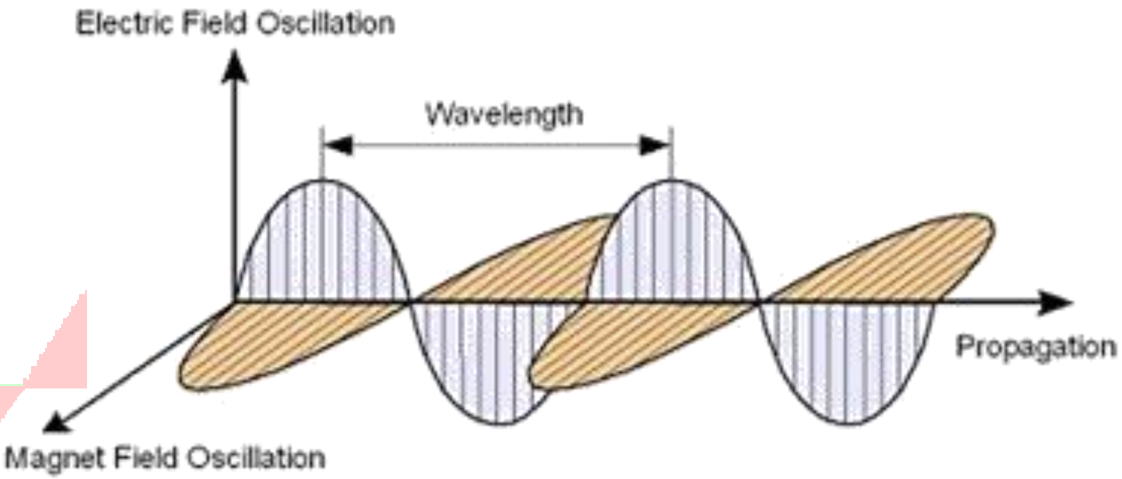
– Quick diagnosis. Eg. Blood blockages.



Exercise 6 and 7
pg. 57-61

HOMEWORK

Electromagnetic Radiation



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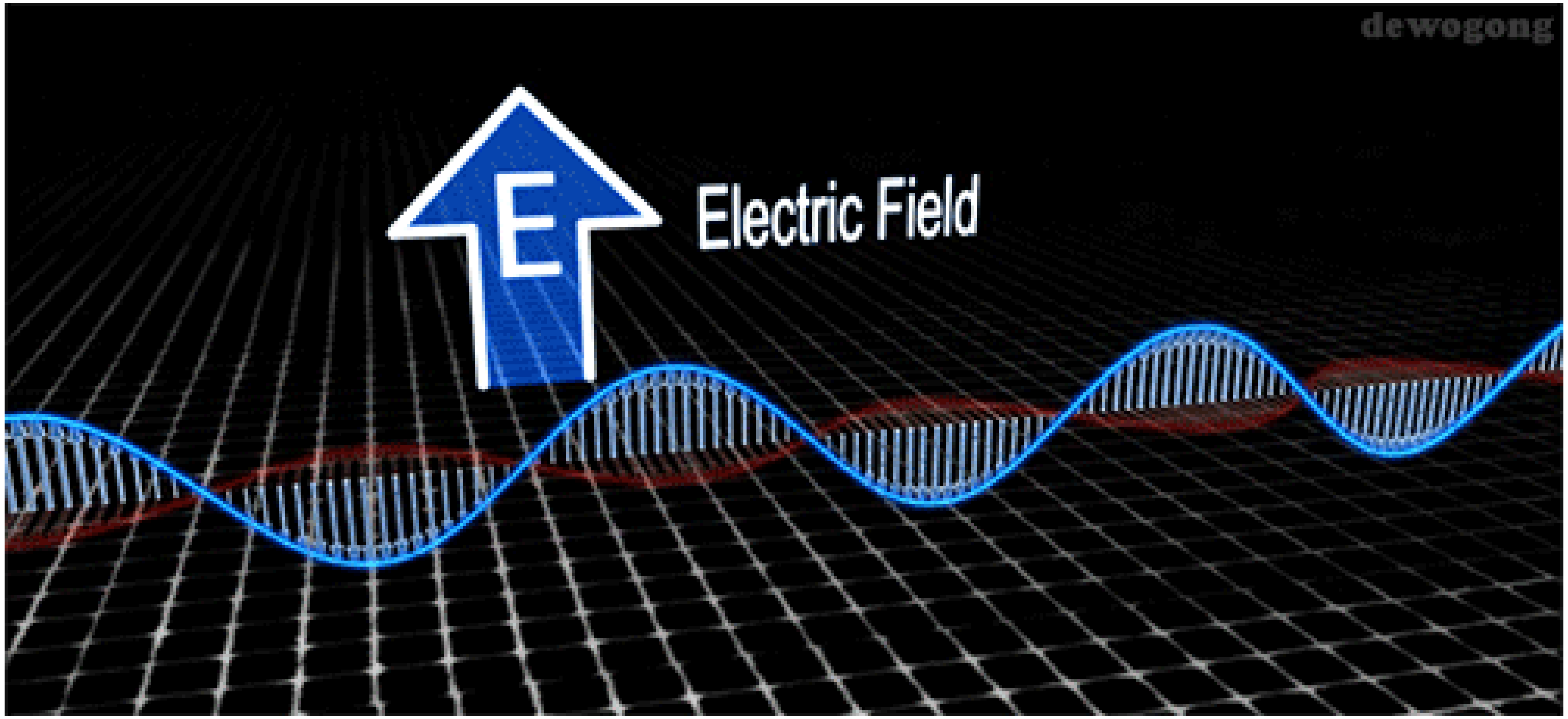
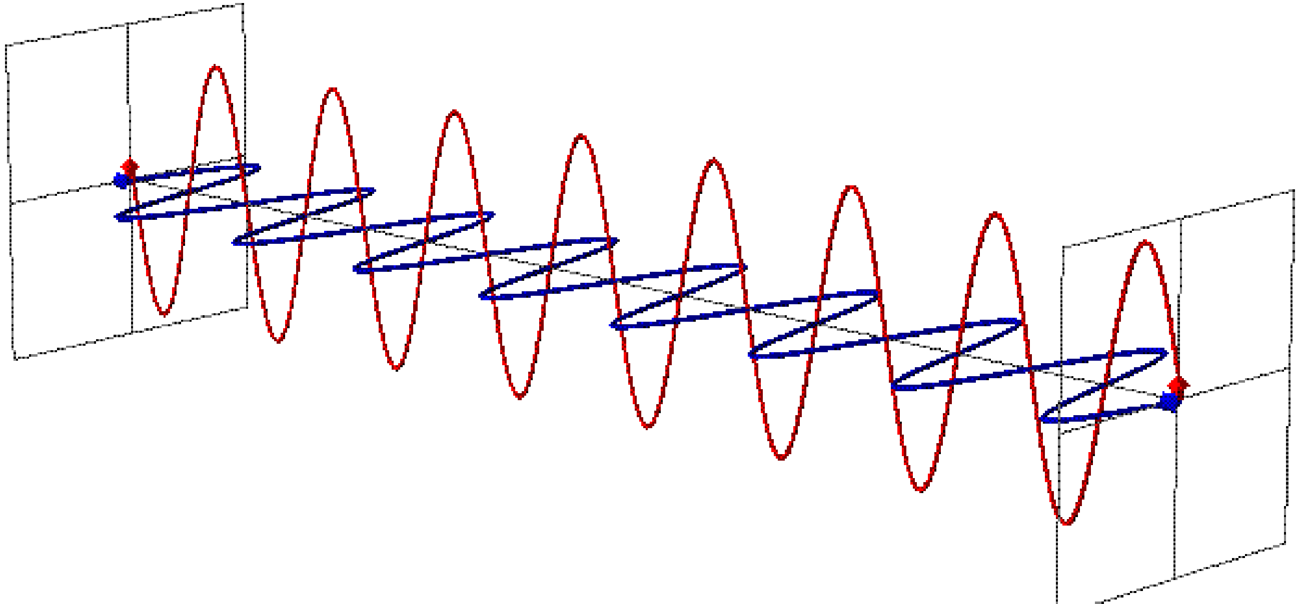
EM **do not** need a medium through which to travel.

The nature of EM waves

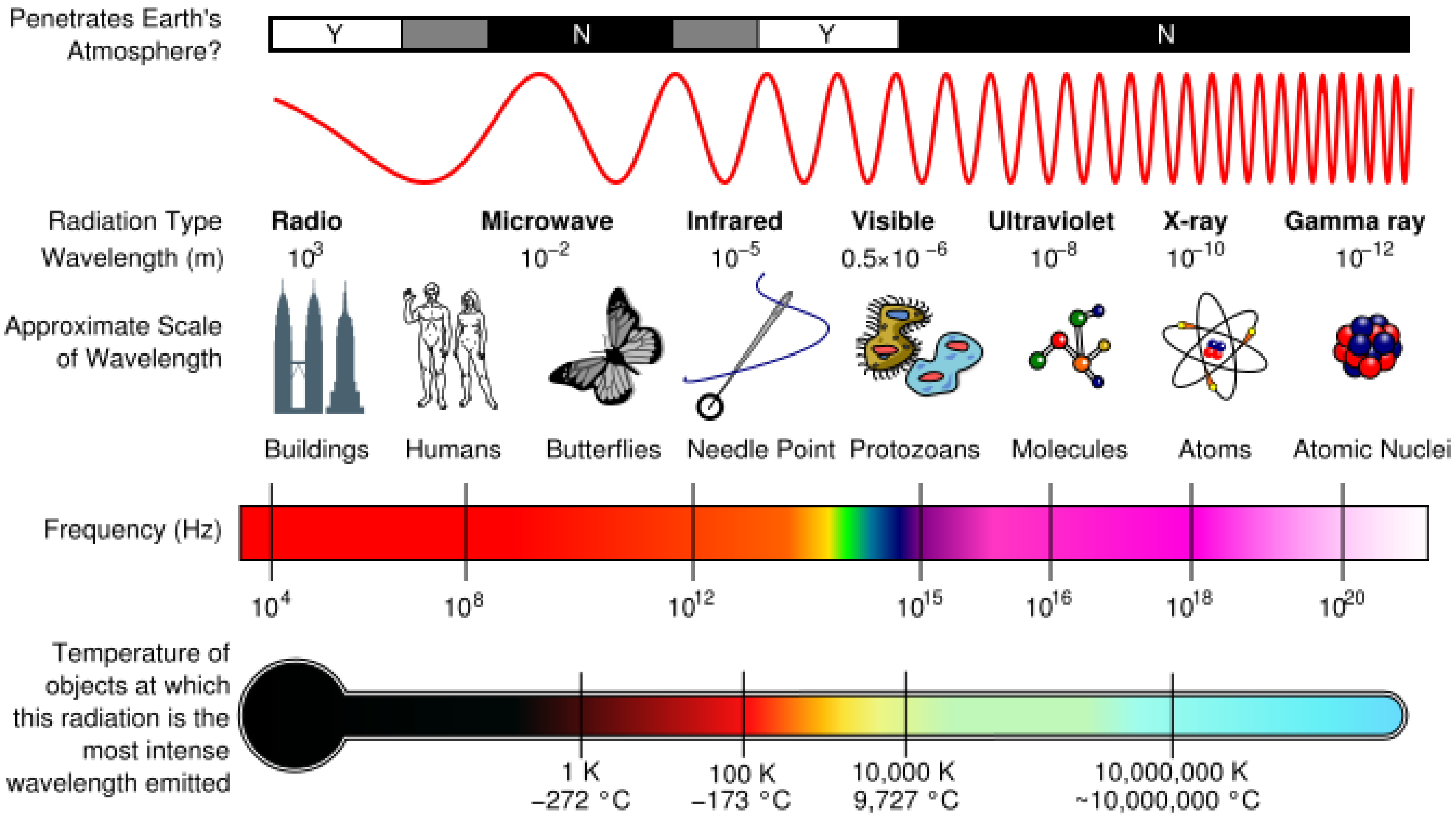
- **1.** Electric **charges** move through the wave.
- **2.** These **accelerating** charges induce a **magnetic field (B)** in the wave that is perpendicular to the direction of their movement.
- **3.** This **changing magnetic field** also induces an **electric field (E)** that is perpendicular to the magnetic field.
- **4.** So an EM wave is one that has **oscillating** magnetic and electric fields that are both perpendicular to each other and to the direction of motion.


dewogong
E
Electric Field



Electromagnetic Spectrum

› Consists of a range of all the different types of EM waves



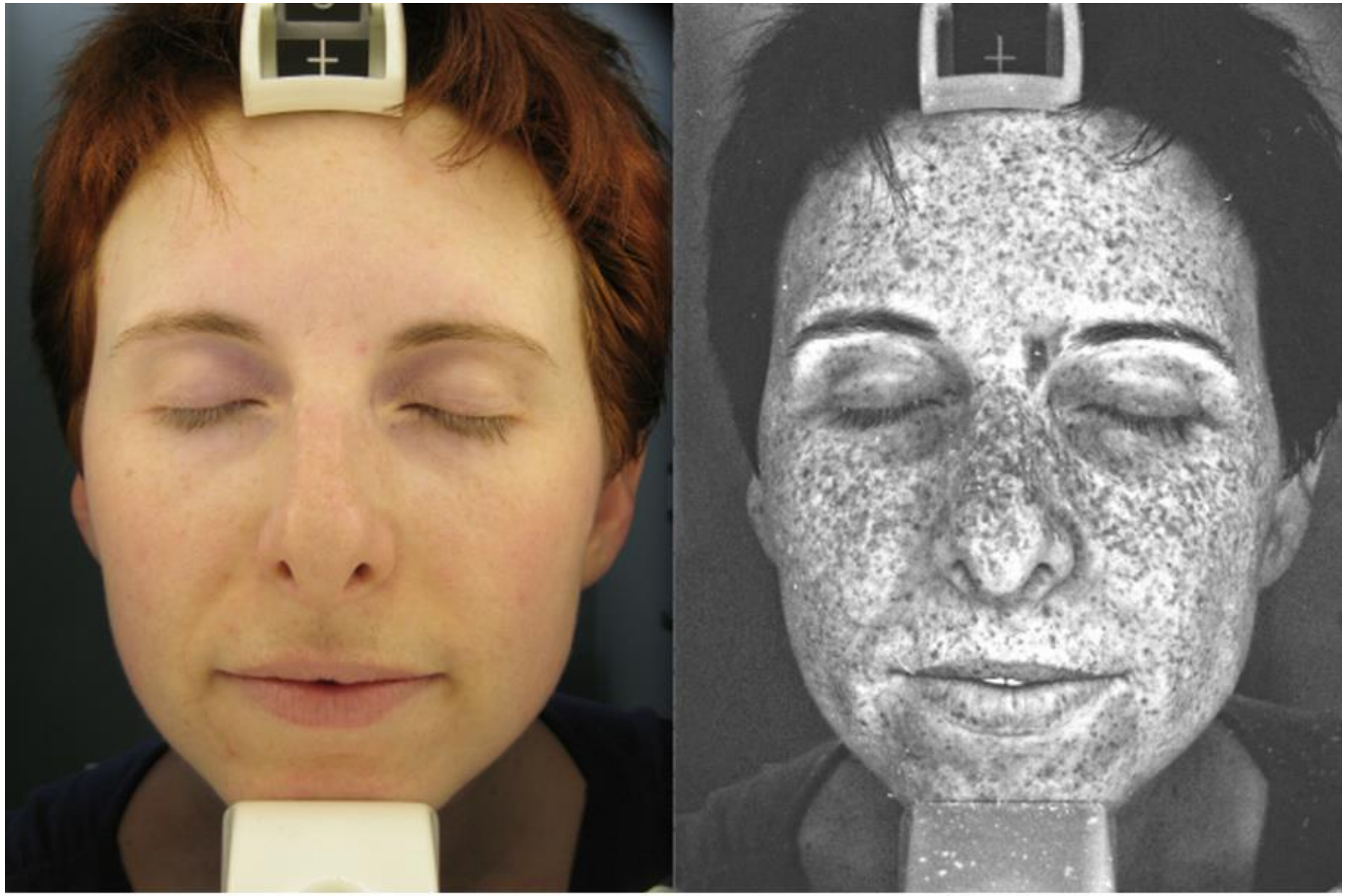
Arranged in order of increasing frequency
and decreasing wavelength

Properties of EM waves

- Moves at a constant **speed** of $3 \times 10^8 m.s^{-1}$.
- $\boldsymbol{v} = f\lambda$ $\quad$ $is\ now$ $\quad$ $\boldsymbol{c} = f\lambda$
- **Do not need a medium** for movement.
- Have all the properties of **waves** – interference, r[illegible]ction and reflection.
- Have **particle** properties.
- **Transverse** waves.

- Sunlight is the **full spectrum** of the electromagnetic radiation produced by the Sun.
- The sunlight is filtered by the earth's atmosphere, and we see the sun's radiation as daylight.

Just for interest…



A 35 Year Old Melanoma Survivor Testing Her Skin Under Normal Light & Ultra-violet Light

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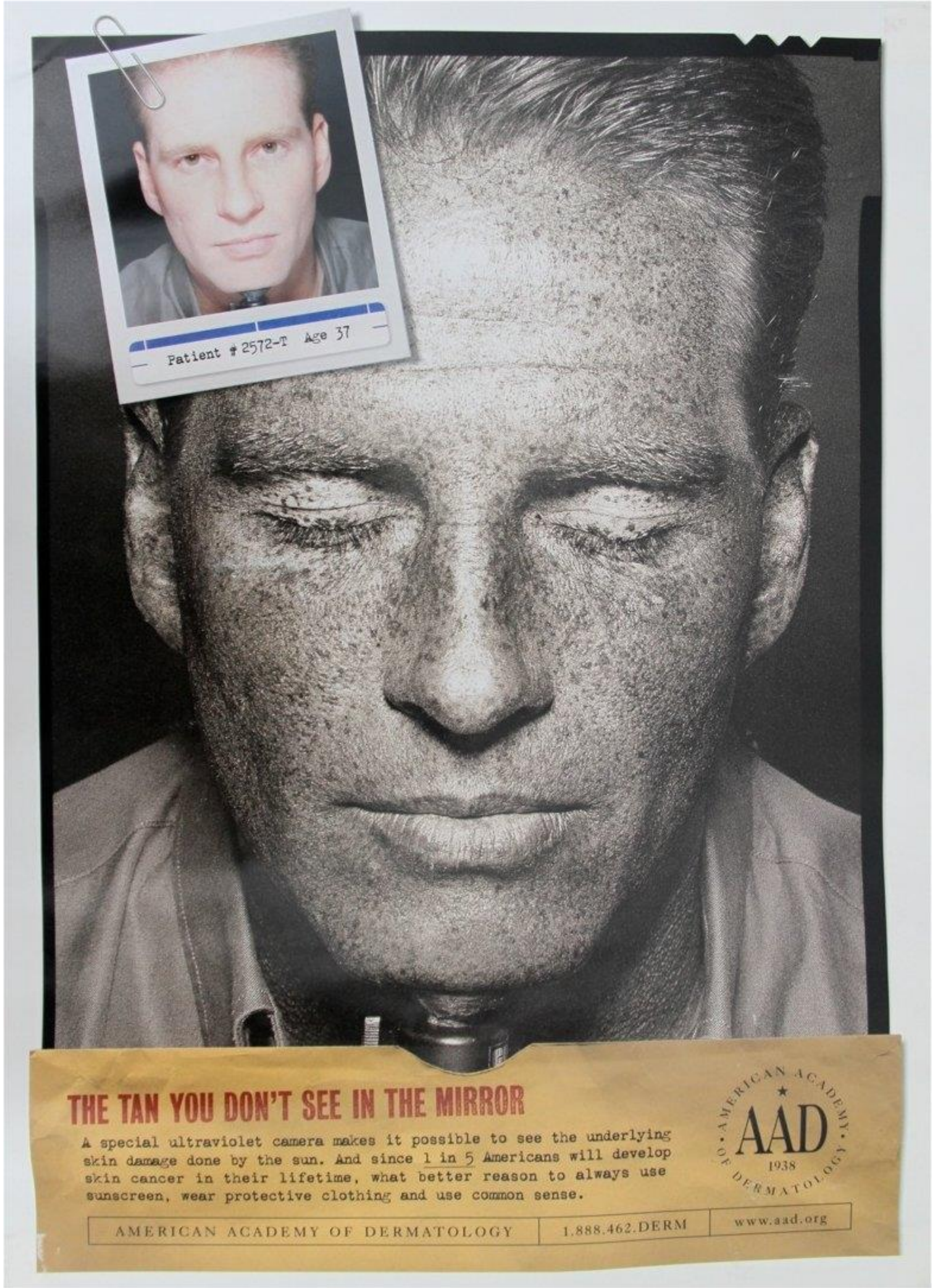
Patient # 2572-T Age 37
THE TAN YOU DON'T SEE IN THE MIRROR
A special ultraviolet camera makes it possible to see the underlying skin damage done by the sun. And since 1 in 5 Americans will develop skin cancer in their lifetime, what better reason to always use sunscreen, wear protective clothing and use common sense.
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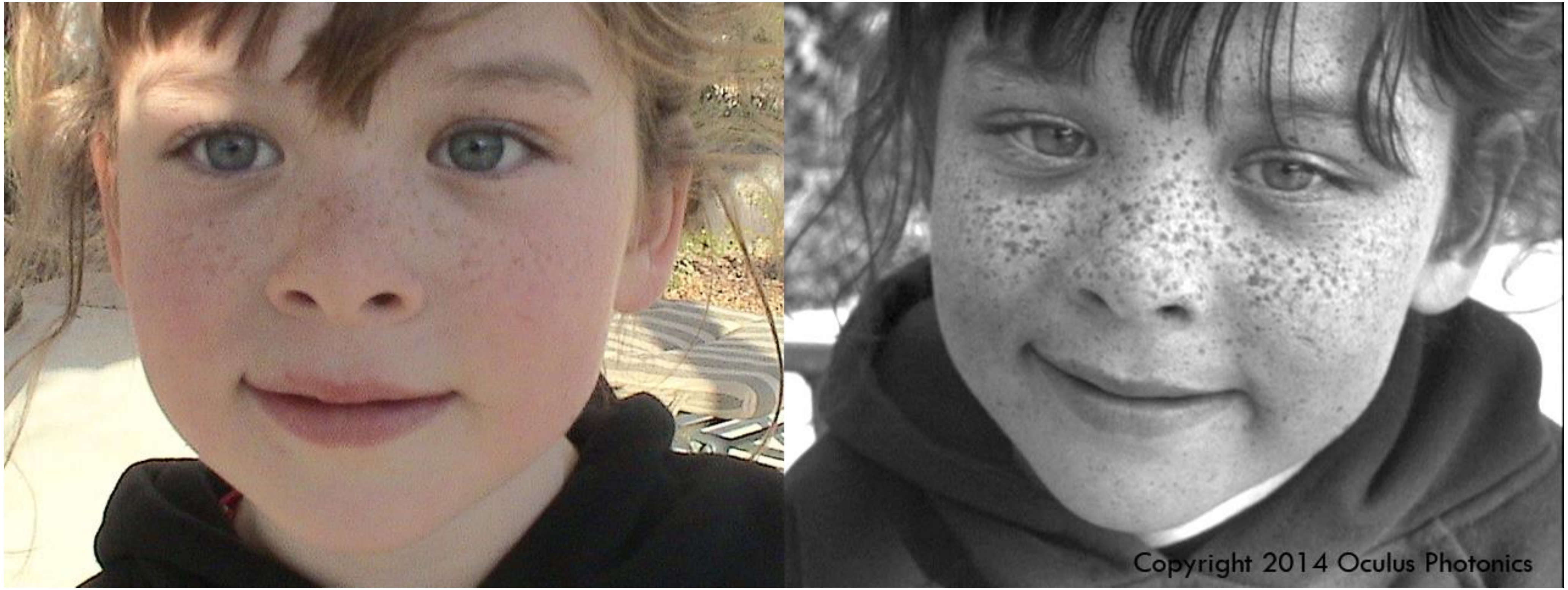
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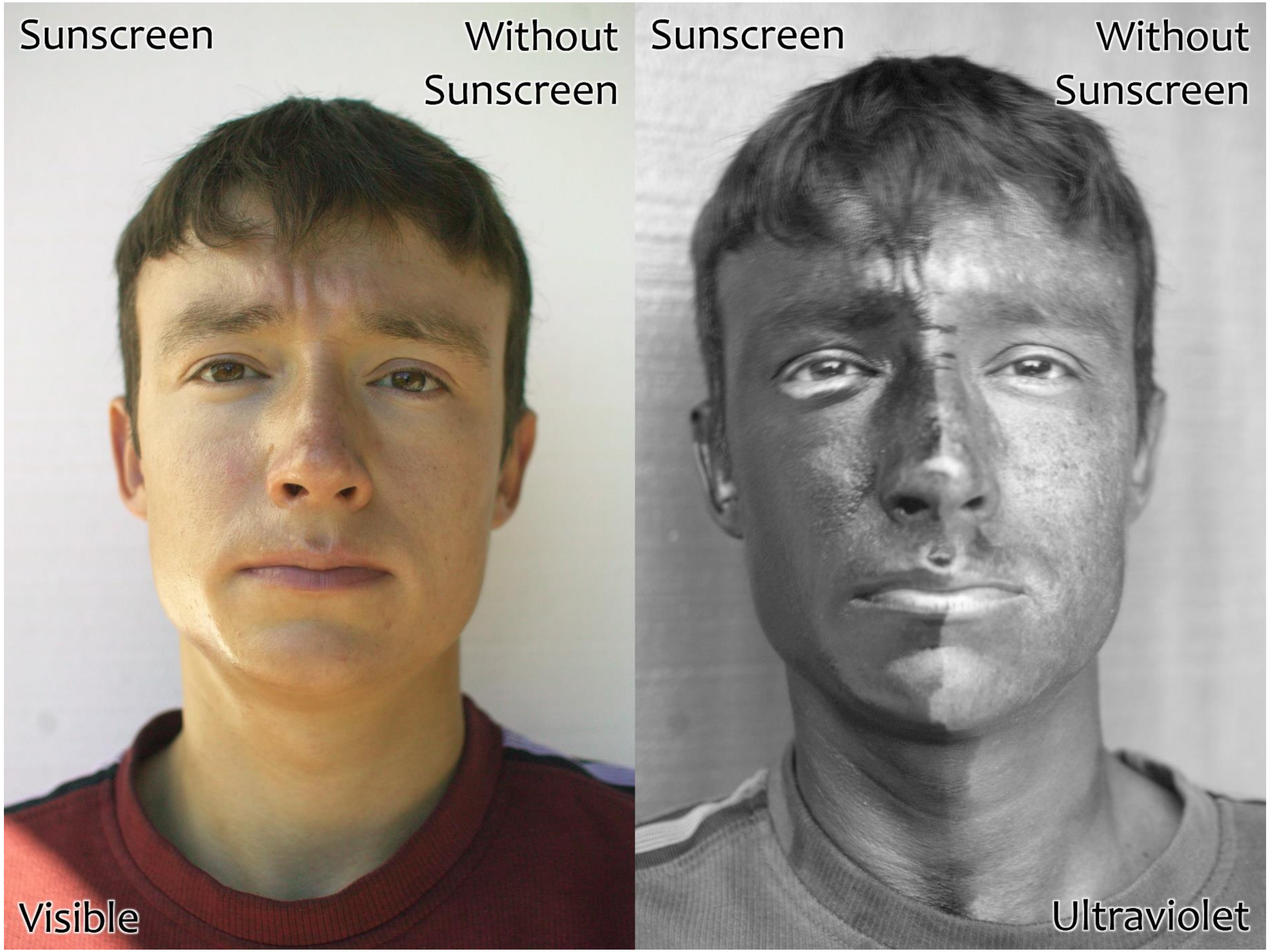

Sunscreen
Without Sunscreen
Visible
Sunscreen
Without Sunscreen
Ultraviolet

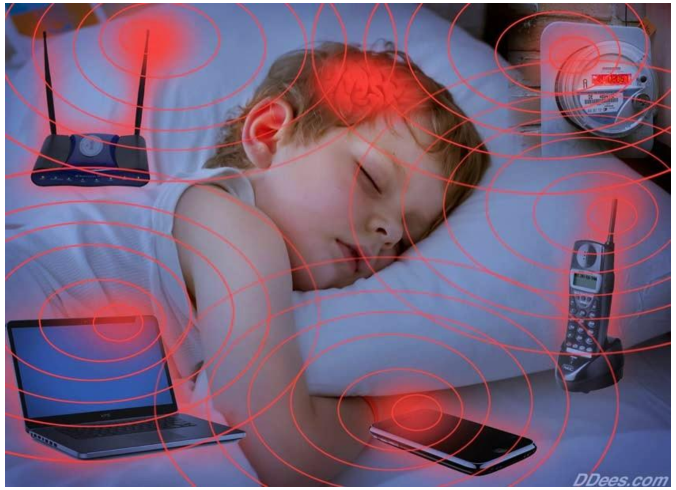


Penetrating ability

- The greater the **energy** of a wave – the greater it's **penetrating ability**.

- Greater **frequency** – greater **penetrating ability**.
- **Gamma rays** (through lead) – **more penetrative than X-rays**. (through soft tissue but not bone).
- UV rays of the sun can travel through clouds on a cloudy day, while infrared rays (which warms your body) cannot.

TYPE OF EM RADIATION	USES	DISADVANTAGES
Radio waves	Radios, TVs, telescopes	Noise pollution
Microwaves	Telephone connections, satellites, cell phones, radar systems, speed traps, microwaves ovens	Use of cell phones can be addictive which leads to decrease in productivity.
Infrared light	Keeping food warm in takeaway restaurants, remote control,	Used by poachers and soldiers for tracking at night.
Visible light	Photosynthesis in plants. Objects reflect light so that we can see them.	
UV light	Fluorescent pens, sterilization of foods	Too much exposure can damage eyes and skin and could cause cancer.
X rays	CT scans. Security scanners, medical images	Too much exposure can lead to cancer and skin damage.
Gamma rays	Radiation of cancer	Released during nuclear reactions. Even result in death.



The particles nature of EM waves

- EM have what we call a "dual nature"
  - Wave properties
  - Particle properties
    - Energy in the wave is transferred in "**packets"** called photons.
    - These photons have a **fixed amount of energy** called "quanta" of energy.

Photon: Energy packets (quanta) that transfer energy to particles of matter.

IMPORTANT CONVERSIONS

- $1mm = 1 \times 10^{-3} m$
- $1\mu m = 1 \times 10^{-6} m$
- $1nm = 1 \times 10^{-9} m$
- $1pm = 1 \times 10^{-12} m$

The energy of a photon can be calculated using:

$$e = hf$$

Frequency (Hz)

Speed of light $(\text{m}.s^{-1})$

BUT

$$e = h\frac{c}{\lambda}$$

energy (J)

Planck's constant $(6.63 \times 10^{-34} J.s)$

$(\text{m}.s^{-1})$

Wavelength (m)

Change in time (s)

EXAMPLE (pg. 70)

1. Infrared rays with a wavelength of 3 μm are released by the Sun. The frequency is $1 \times 10^{14} Hz$. Calculate how much energy the infrared photons have.

2. Calculate the energy of a photon of violet light with a wavelength of 410nm.

3. A photon of infrared light has $1\,199 \times 10^{-20} J$ of energy. Calculate the frequency of the infrared light.

4. A photon of a microwave has $3.44 \times 10^{-23} J$ of energy. Calculate the wavelength of the microwave.

Exercise 8 pg. 71-74

HOMEWORK

Test Examples